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Optimisation of Low-cost PID Microcontroller Based Syringe Pump Platform for Physiological Pressure Waveform Replication

BY

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Abstract

This project presents a low-cost, microcontroller-based platform capable of reproducing physiological pressure waveforms using a NE-1010 syringe pump actuated through closed-loop PID control. An Arduino Mega, paired with a medical-grade pressure transducer and ADC amplifier, was used to achieve real-time pressure feedback and waveform tracking. The system successfully generated physiologically relevant pressure profiles, with performance influenced by pump dynamics, outlet resistance, and sensor sampling limitations. Despite these constraints, results demonstrate that accurate and repeatable pressure regulation is feasible with inexpensive hardware, offering an accessible alternative to high-cost laboratory systems and enabling broader adoption in biomedical research environments.

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Table of Contents

Abstract iii
Acknowledgements iv
Table of Contents v
List of figures vii
List of tables viii
Nomenclature ix
Introduction 1
1. Literature review 2
1.1 Fluid Dynamics and Haemodynamics 2
1.2 Importance of Vessel Geometry 4
1.2.1 Geometry and Fluid Resistance in Physical Models 4
1.2.2 Programmable and microfluidic Geometries 8
1.3 Fluid Pressure Control Platforms 9
1.4 Arduino Based Sensor Adaptation 17
1.5 Literature Review Conclusion 17
2 Methodology 18
2.1 Pressure Sensor Calibration and Verification. 18
2.1.1 Signal Conditioning 19
2.1.2 Calibration Procedure 20
2.1.3 Sensor Noise, Drift, and Filtering 21
2.2 Liquid Media Dispensing System 22
2.3 Hydraulic resistance and laminar flow 23
2.4 Final System Configuration and Cost 26
2.5 Control Architecture 28
2.5.1 PID Tuning Approach – Ziegler-Nichols Method 28
2.5.2 Closed-Loop PID Evaluation Using Custom Input Pressure Waveforms 30
2.6 Physiological Pressure Waveform Replication 31
3 Results and discussion 33
3.1 Pressure sensor stability 33
3.2 Hydraulic Resistance Results 35
3.3 Custom Input Waveform Reproduction Results 38
3.4 Physiological Waveform Reproduction Results 40
3.5 Discussion 45

4 Conclusions 48
5 References 49

List of figures

FIGURE 1 FLUID RESISTIVE VESSEL 5
FIGURE 2 LAKE ET AL. (2017) FLOW RESISTANCE 6
FIGURE 3PATIL ET AL. (2015) EXPERIMENTAL SETUP 6
FIGURE 4 PATIL ET AL. (2015) RESULTS: (RED) – PLEXUFIX 24 G × 50 MM; (BLACK) – STIMUPLEX A 22 G × 50 MM; (GREEN) – STIMUPLEX A 21 G × 100 MM; (PINK) – STIMUPLEX A 20 G × 150 MM; (CYAN) – CONTIPLEX D 18 G × 55 MM; (RED DOTTED) – TUOHY 18 G × 50 MM; (GREEN DOTTED) – PAJUNK 19 G × 10 MM; (BLACK DASHED) – 75 KPA THRESHOLD LINE 7
FIGURE 5 LEE ET AL. (2022) RESULTS 22 GAUGE NEEDLE PRESSURE-FLOWRATE 7
FIGURE 6 VIJAYARATNAM ET AL. (2021) EXPERIMENTAL SETUP 10
FIGURE 7 PATIENT PRESSURE WAVEFORM ITERATIVE RESULTS 11
FIGURE 8 HELMS ET AL. (2021) EXPERIMENT SETUP 12
FIGURE 9 HELMS ET AL (2021) RESULTS 13
FIGURE 10 LAKE ET AL.(2017) EXPERIMENTAL SETUP 14
FIGURE 11 LAKE ET AL. (2017) RESULTS 14
FIGURE 12 WATSON AND SENYO (2019) EXPERIMENT SETUP 16
FIGURE 13WATSON AND SENYO (2019) RESULTS 16
FIGURE 14 DI DOMENICO (2020), SENSOR CALIBRATION RESULTS 17
FIGURE 15 IN VITRO PID CONTROL ARCHITECTURE 18
FIGURE 16 (A) HX711 CIRCUIT DIAGRAM (B) HX711 ADC AMPLIFIER - PHYSICAL MODEL DISPLAYING 10 AND 80 HZ ROUTING PADS 20
FIGURE 17 (A) UTAH MEDICAL DELTRAN TRANSDUCER SAMPLE CALIBRATION DATA (B) DELTRAN TRANSDUCER WATER COLUMN CALIBRATION SETUP 21
FIGURE 18 (A) ARDUINO MEGA CONNECTION SETUP WITH NE-1010 SYRINGE PUMP (B) NULL MODEM CABLE CONFIGURATION (C)RS-232 ARDUINO MODULE 23
FIGURE 19 IN VITRO OUTLET CONFIGURATION - (A) 0.6 MM ID NEEDLE, (B) TEE ADJUSTABLE VALVE, (C) 0.5 MM ID TUBING + NEEDLE 24
FIGURE 20 FINAL EXPERIMENTAL SETUP 26
FIGURE 21 (A) EXPERIMENTAL ASSEMBLY (B) HYDRAULIC VESSEL CONFIGURATION 27
FIGURE 22 ZIEGLER-NICHOLS EXPERIMENTAL RESPONSE 29
FIGURE 23 CUSTOM CONTROLLED INPUT PROFILES - (A) TRAPEZOIDAL (B) SQUARE (C) CIRCULAR (D) TRIANGULAR 31
FIGURE 24 ORIGINAL PHYSIOLOGICAL WAVEFORMS (SOURCED FROM WILLEMET ET AL (2015)) – ALL DATA POINTS ARE SHIFT TO A BASELINE PRESSURE OF 80 MMHG 31
FIGURE 25 UTAH MEDICAL DELTRAN STATIC PRESSURE DATA 34
FIGURE 26 PRESSURE VS FLOWRATE MEASURE AT 8,12,16,20 ML/MIN FOR THREE OUTLET CONFIGURATIONS - (A) CONTROLLABLE VALVE, (B) 0.6 MM ID NEEDLE, (C) 0.5 MM ID TUBING + NEEDLE 35
FIGURE 27 0.6 MM ID NEEDLE PRESSURE DROP RESULTS 37
FIGURE 28 NE-1010 COMPARISON RESULTS AT 4, 20 ML/MIN, OLD MODEL (1), NEW MODEL (2) 37
FIGURE 29 PRESSURE OSCILLATION FOR IMPROVED PUMP AND 0.6 ID MM NEEDLE OUTLET CONFIGURATION 38
FIGURE 30 CONTROLLED INPUT WAVEFORM RESULTS 39
FIGURE 31 CONTROLLED INPUT WAVEFORM RESULTS - ZOOM INTO PEAK END 40
FIGURE 32 PHYSIOLOGICAL WAVEFORM REPRODUCTION FOR CAROTID CYCLES 41
FIGURE 33 PHYSIOLOGICAL WAVEFORM REPRODUCTION FOR AORTIC ROOT CYCLES 42
FIGURE 34 PHYSIOLOGICAL WAVEFORM REPRODUCTION FOR ILIAC CYCLES 42
FIGURE 35 SUPERIMPOSED WAVEFORMS TO DIGITAL WAVEFORMS 43
FIGURE 36 ERROR OF IN VITRO WAVEFORM TO DIGITAL WAVEFORM (SUPERIMPOSED) 44

List of tables

TABLE 1REYNOLD'S NUMBER OUTLET ANALYSIS.....25
TABLE 2 ZIEGLER-NICHOL'S CONTROLLER GAIN PARAMETERS.....30
TABLE 3UTAH MEDICAL DELTRA STATIC PRESSURE ANALYSIS (MEAN VALUE, STD MMHG, MAX ERROR BETWEEN MEAN AND MEASURED, MIN ERROR).....34
TABLE 4 OUTLET CONFIGURATION RESULTS.....36
TABLE 5 MODEL 1 VS MODEL 2 FREQUENCY RESULTS.....38
TABLE 6 BEST PERFORMING SUPERIMPOSED WAVEFORM RESULTS.....45

Nomenclature

A	=	Amplitude
cmH^2O	=	Centimetres of water
D	=	Diameter
$e[i]$	=	Discrete time error
E	=	Young's modulus
g	=	Gravitational acceleration
h	=	height
Hz	=	Hertz (frequency)
K	=	Process Gain
K_p	=	Proportional gain
$K_P^{discrete}$	=	Discrete proportional gain
$K_i^{discrete}$	=	Discrete integral gain
$K_d^{discrete}$	=	Discrete derivative gain
L	=	System Dead Time
mmHg	=	Millimetres of mercury
m	=	Samples
r	=	Vessel radius
R	=	Resistance
Re	=	Reynolds number
T	=	Time Constant (Ziegler-Nichol's Method)
T_I	=	Integral time
T_D	=	Derivative time
$u[n]$	=	Discrete time control output
V	=	Volts
v	=	Fluid velocity
ΔP	=	Pressure change
Δt	=	Time change (period)
Δu	=	Input change
Δy	=	Output change
μV	=	Microvolts

Q	=	Volumetric flowrate
α	=	Womersley number
μ	=	Dynamic viscosity
π	=	Pi
ρ	=	Fluid density
ω	=	Angular frequency
h	=	Wall thickness

Introduction

Accurate reproduction of physiological pressure waveforms is essential in biomedical research, supporting applications such as medical-device testing, drug-delivery validation, and studies of complex biological processes. Existing laboratory systems capable of generating these waveforms are typically complex, costly, and large-scale, making them inaccessible for smaller educational research environments. Conversely, low-cost microcontroller-based systems are generally limited by poor pressure fidelity and insufficient high-frequency response times, preventing them from accurately reproducing physiological dynamics. Developing an affordable, reconfigurable platform capable of generating patient-specific pressure profiles would therefore significantly expand experimental capability, particularly in low-resource settings.

This thesis presents the development of a low-cost, microcontroller-based platform capable of replicating physiologically realistic pressure waveforms in real time. The system brings together an Arduino Mega for control and computation, a medical-grade Utah Medical Deltran pressure transducer with an HX711 ADC amplifier for high-resolution sampling, and a New Era NE-1010 syringe pump for linear flow actuation. Together, these subsystems form an integrated, compact, and reconfigurable closed-loop control platform designed specifically for biomedical pressure waveform reproduction. A proportional–integral–derivative (PID) controller continuously modulates the pump's flow rate based on instantaneous pressure feedback, enabling the system to follow complex physiological waveforms with high fidelity. Beyond demonstrating the feasibility of achieving closed-loop pressure control using accessible hardware, the study identifies the underlying mechanical, sensing, and control factors that ultimately determine system performance, providing a foundation for future enhancements and broader biomedical applications.

1.Literature review

1.1 Fluid Dynamics and Haemodynamics

Before attempting to replicate any physiological process in vitro, it is essential to first understand the underlying mechanics that govern it. In the context of cardiovascular dynamics, this begins with a detailed understanding of fluid motion, vessel behaviour, and the principles that shape pressure and flow throughout the circulatory system. These haemodynamic foundations determine how real vessels respond to pulsatile inputs, how pressure waves propagate, and how resistance, compliance, and inertance interact to produce characteristic waveform features.

To understand the physical nature of blood flow itself. Secomb (2016) offers a comprehensive overview of the governing principles of haemodynamics, describing how blood pressure and flow are regulated by a dynamic interplay between cardiac output, vessel compliance, and vascular geometry. Most notably, Secomb highlights the critical role of pressure gradients in driving flow, and how these are influenced by parameters such as vessel radius, viscosity, and length, as formalised in Poiseuille's Law:

$$\Delta P = \frac{8\mu L Q}{\pi r^4} \tag{1}$$

This equation expresses Poiseuille's law, where the pressure drop ΔP is determined by the fluid viscosity μ, the tube length L, and the flowrate Q, and depends inversely on the fourth power of the vessel radius r.

This relationship illustrates how even minor changes in vessel radius which can occur naturally in pulsatile systems can drastically alter flow characteristics. In our context, this validates the need for real-time pressure regulation within the syringe pump system, especially when replicating waveforms in compliant media like elastomeric tubing or engineered vessel models. Secomb also identifies more complex phenomena such as pulse wave propagation, driven by the viscoelastic properties of arterial walls. These effects are described by the Moens-Korteweg equation:

$$c = \sqrt{\frac{Eh}{2\rho R}} \tag{2}$$

The equation shows that wave speed c increases with the vessel's elastic modulus E and wall thickness h, and decreases with the fluid density ρ and vessel radius R, which in turn affects timing and morphology

of the pressure waveform. These dependencies are particularly important when trying to simulate arterial conditions such as hypertension or aging, where vessel compliance changes significantly. For our project, understanding this relationship helps inform on the expected pressure response time and rise/fall dynamics required of the control loop. Another important parameter is the Womersley number α, which captures how flow velocity profiles shift under pulsatile (oscillatory) conditions:

$$\alpha = R\sqrt{\frac{\omega\rho}{\mu}} \tag{3}$$

Where, R is the radius, ω is the oscillation frequency, ρ is the fluid density, μ and is the dynamic viscosity.

Building on these theoretical principles, Thomas and Sumam (2016) review a broad range of numerical and experimental approaches to simulating arterial blood flow. They evaluate models that incorporate non-Newtonian blood behaviour, fluid–structure interaction (FSI), and compliant vessel geometries, noting how assumptions such as rigid walls or Newtonian fluid properties often oversimplify the true system. Their review categorises simulation approaches by dimensionality (1D vs 3D), vessel wall characteristics (elastic vs viscoelastic), and mathematical frameworks (e.g., linearised Navier-Stokes, CFD, FVM/FEM). Within this discussion, they further emphasise the role of laminar flow assumptions, Reynolds number regimes, and vessel geometry in shaping predicted pressure and velocity fields. The Reynolds number (Re) is introduced as a dimensionless ratio of inertial to viscous forces, used to differentiate laminar from transitional or turbulent flow; its typically modest values in arterial circulation reflect the predominance of viscosity-driven motion. Vessel geometry is likewise identified as a central determinant of haemodynamics, with parameters such as diameter, curvature, tapering, and wall thickness governing local resistance, wave-propagation behaviour, and the distribution of shear stresses along the arterial wall. Together, these considerations illustrate the inherent trade-offs between physiological realism and computational cost.

While their analysis affirms the theoretical potential of computational haemodynamics, it also reveals practical limitations, namely, that real-time simulation remains largely unattainable due to computational demand required for modelling physiologically accurate models. This presents a key opportunity for physical emulation systems like the one proposed in this thesis, rather than simulating flow digitally.

The limitations presented by Thomas and Suman (2016) are further emphasises by the work of Menon et al.(2023), who demonstrate the practical consequences of these fluid-dynamic constraints within

personalised coronary flow modelling. Their work integrates clinical CT perfusion imaging with synthetic vascular tree generation, enabling full heart simulation of myocardial and coronary blood flow. Most notably, they demonstrate that conventional boundary-condition rules such as Murray's law fail to capture real flow distributions, with Menon et al. reporting errors exceeding 600% when empirical diameter-based scaling was used in place of personalised perfusion data. These discrepancies highlight the sensitivity of coronary haemodynamics to geometric fidelity, microvascular resistance, and the nonlinear pressure–flow relationships that emerge in branching networks.

Drawing from these insights, it becomes clear that pressure waveform reproduction must be tailored to the specific mechanical and physiological context of the simulated vessel environment. Systems must be able to adapt to downstream variations in compliance and resistance, while also supporting multiple waveform types, such as aortic, brachial, or pathological profiles. Menon et al.'s (2023) emphasis on hybrid modelling combining global control (e.g., pump force) with local tuning (e.g., microvascular pressure targets) exemplifies the level of control required to achieve physiological haemodynamics. Their work highlights the importance of a well-defined control framework that aligns both theorical and practical characteristics. This principle forms the conceptual basis for the present project, which seeks to reproduce realistic pressure waveforms and adapt to differing downstream conditions through an accessible, experimentally driven platform.

1.2 Importance of Vessel Geometry

Having established the principles that govern blood flow, it is equally important to consider how the geometry and mechanical properties of blood vessels fundamentally govern how pressure and blood flow behave through the circulatory system. Understanding these dependencies is crucial for trying to replicate an environment that has the capabilities to generate physiologically accurate pressure behaviours.

1.2.1 Geometry and Fluid Resistance in Physical Models

Vessel geometry determines the spatial distribution of flow and the magnitude of local pressure gradients. Even minor reductions in radius, curvature, or tapering can cause disproportionate increases in resistance and wave-speed variation, influencing the morphology of the pressure waveform. Thomas and Sumam (2016) highlight that such geometric and elastic factors control the attenuation and reflection of pressure waves along arteries, and that simplified rigid-wall assumptions can misrepresent true pulse behaviour.

In experimental settings, this means that tube diameter and material stiffness must be chosen to emulate the compliance and damping of real arteries, ensuring that rise and fall times remain physiologically realistic. In simplified benchtop loops, compliance is primarily governed by tubing elasticity: stiffer materials yield faster transients and greater reflection of pressure waves, while more compliant silicone lines introduce damping and delay. Careful selection of these parameters allows the vessel assembly to function as a tuneable analogue for arterial elasticity.

An illustrative example of this is provided by Lake et al. (2017), who developed a low-cost pressure platform incorporating feedback control for microfluidic applications. Using a simple micro-channel geometry; a linear channel and a Y-junction , where they compared three operational modes: a single commercial syringe pump, two independent open-loop custom pumps, and two feedback-controlled pumps. The first two setups produced unsteady pressure and periodic drift of the laminar interface within the Y-junction (Figure 1), confirming that open-loop control could not maintain equal inlet conditions. When pressure feedback was introduced through PID controller, the interface became stable and centred, with steady pressures held within 1% of the setpoint and ~1 second response time.

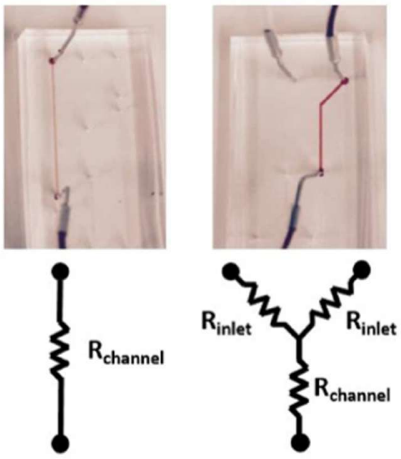


Figure 1 Fluid Resistive Vessel

Their results highlight how geometric restriction and feedback control jointly determine pressure stability in fluidic systems. The study shows that channel geometry can be used as a designed resistive element to shape hydraulic behaviour, and that active pressure regulation can mitigate instabilities caused by unequal or time-varying flow resistance (Figure 2). This finding is directly relevant to the design of any pressure-driven flow system where tubing dimensions, outlet resistance, or branching geometries influence dynamic response.

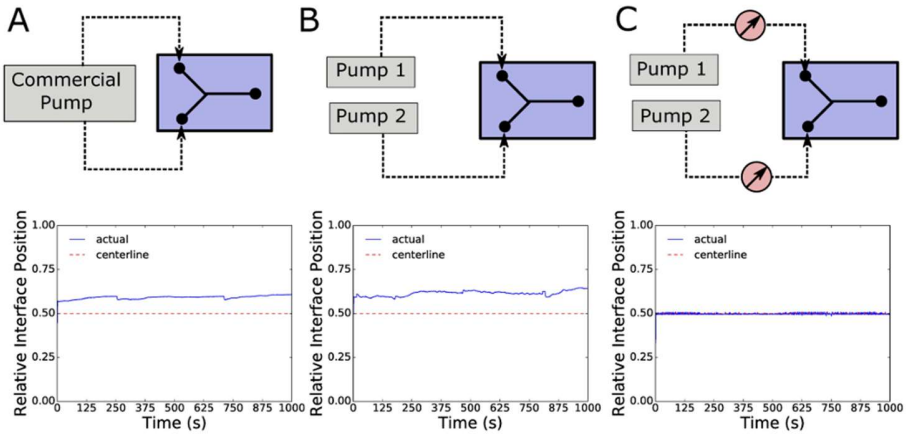


Figure 2 Lake et al. (2017) Flow Resistance

Patil et al. (2015) conducted a controlled bench-top investigation to quantify how needle gauge, length, and infusion rate affect upstream injection pressure in syringe-driven systems. Their experimental setup (Figure 3) consisted of a clinical syringe pump delivering saline through seven commonly used regional-anaesthesia needle systems, with needle outlets discharging directly to open air. A fluid-filled pressure transducer, positioned between the syringe and needle, recorded real-time pressure traces through a 240 Hz ADC and proprietary waveform-analysis software. By isolating the needle resistance from any downstream tissue or fluidic load, the study was able to characterise the pure hydraulic contribution of needle geometry.

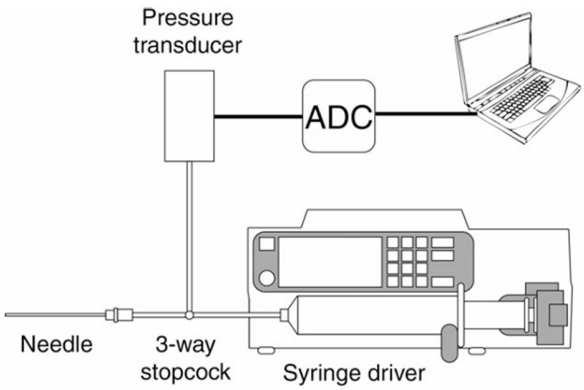


Figure 3Patil et al. (2015) Experimental Setup

The results from Patil et al. (2015) clearly demonstrate that injection pressure increases almost linearly with flowrate for every needle type tested. Across all seven regional anaesthesia needle systems, plateau pressures showed a strong proportional relationship to the imposed flow rates of 5–20 mL/min (Figure 4). This linearity held consistently across all gauges, with low standard deviations indicating high

repeatability of the measurements. As expected, reductions in outlet geometry produced markedly higher pressure for the same flowrate, reflecting the dominant role of geometric restrictions in defining hydraulic resistances.

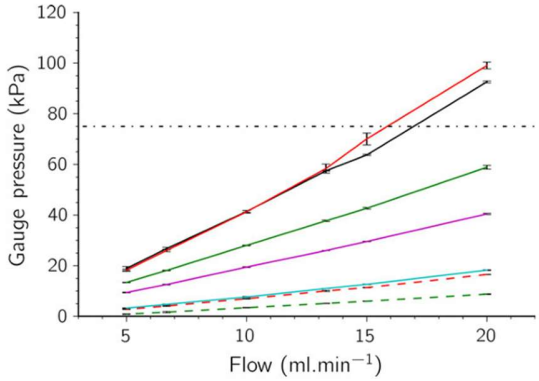


Figure 4 Patil et al. (2015) Results: (red) – Plexufix 24 G × 50 mm; (black) – Stimuplex A 22 G × 50 mm; (green) – Stimuplex A 21 G × 100 mm; (pink) – Stimuplex A 20 G × 150 mm; (cyan) – Contiplex D 18 G × 55 mm; (red dotted) – Tuohy 18 G × 50 mm; (green dotted) – Pajunk 19 G × 10 mm; (black dashed) – 75 kPa threshold line

Lee et al. (2022) investigated syringe-driven needle systems using an outlet analytical model that predicts the pressure exerted by an exiting saline stream on a wound surface. Unlike Patil et al's pressure studies, which measure the internal hydraulic pressure upstream of a restriction, this analysis calculates the external dynamic pressure generated as fluid leaves the needle. Pressure at the needle tip was assumed to be atmospheric, and the authors computed the downstream impact pressure from the jet velocity using Bernoulli's equation, resulting in a purely analytical estimate rather than a measured internal pressure value. Owing to this modelling choice, the predicted pressure–flow relationship is inherently quadratic (Figure 5), as the impact pressure scales with the square of velocity ($p \propto V^2$) and therefore accelerates rapidly at higher flowrates.

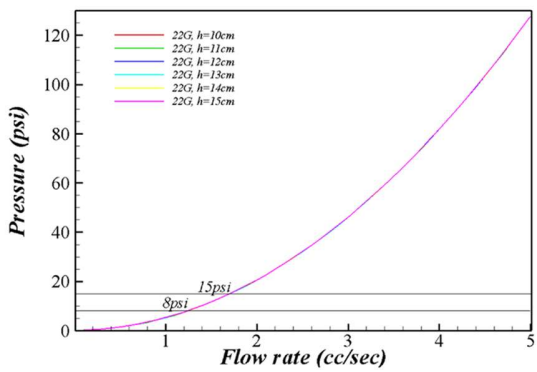


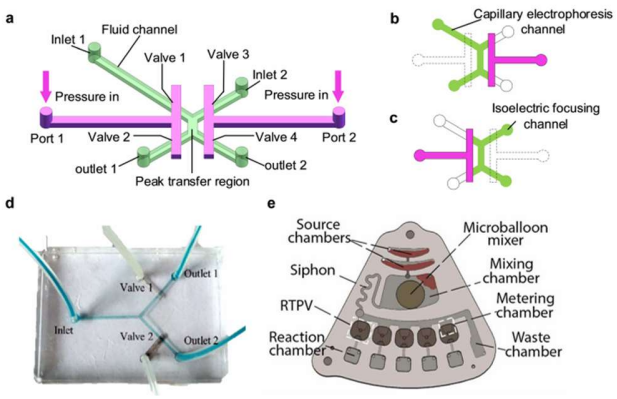
Figure 5 Lee et al. (2022) Results 22 gauge needle pressure-flowrate

Although this differs fundamentally from the linear laminar pressure–flow relationship observed inside closed systems such as those tested by Patil et al. (2015), the findings remain informative for understanding fluid dynamic behaviour as liquid exits a confined vessel. The pronounced exponential-like rise in outlet pressure highlights the increasing role of inertial forces once flow accelerates beyond the internal laminar regime, illustrating how a narrow outlet can transform steady internal pressure into a rapidly intensifying jet. This behaviour reinforces a core design consideration: geometry at the vessel exit strongly influences both internal hydraulic resistance and the magnitude of the downstream pressure impulse, particularly in systems transitioning from laminar internal flow to free-jet external flow.

These insights from Patil et al. (2015), Lee et al (2022), Lake et al (2017) are valuable for the design of pressure-driven flow platforms, as they demonstrate how simple geometric parameters, such as; internal diameter of a vessel and outlet configuration can establish predictable resistance profiles without requiring active flow control. However, the non-linear, quadratic dependence between pressure and flow identified by Lee et al. (2022) also highlights potential limitation of such passive systems, since this behaviour may introduce non-linearity into any feedback-controlled regulation scheme. However, it establishes the need for careful model characterisation. Translating these findings to physical modelling, downstream components must therefore be treated as active elements that shape overall system dynamics rather than passive system parameter.

1.2.2 Programmable and microfluidic Geometries

Beyond passive resistive elements, programmable microfluidic architectures offer a fundamentally different approach to controlling hydraulic behaviour. Rather than relying on fixed tubing dimensions, these systems incorporate micro-valves, deformable channels, and reconfigurable structures that can dynamically alter the cross-section of a flow path. As reviewed by Song et al. (2024), actuation mechanisms such as pneumatic membranes, electrokinetic valves, electrowetting, thermo-responsive hydrogels, and digital microfluidics enable fine control over local resistance and flow distribution. By making geometry a programmable variable rather than a static constraint, these devices can replicate physiologically relevant behaviours such as time-varying vascular resistance, local constriction and dilation, or selective routing through branching networks.



From a design standpoint, these systems provide clear advantages. Offering higher real-time control over downstream vessel fluid dynamics through imposed geometric changes. Programmable restrictions can emulate events that are otherwise difficult to reproduce with a static vessel, such as dynamic stenosis, pulsatile resistance changes, or localised compliance modulation. Their capacity for real-time configuration also make them a valuable conceptual model for understanding how pressure waveforms behaviour at a more physiologically accurate scale.

However, Song et al. (2024) also identifies several practical limitations that are important when considering low-budget platform. Many actuation methods require external pneumatic supplies, high voltage drivers, magnetic fields, optical components, or multi-layer fabrication processes, which increase the cost and system complexity. Response times vary widely between the mentioned actuation methods, electrokinetic valves may respond within seconds compared to hydrogel bases vessels, restricting their use in rapid waveform modulation. Furthermore, some micro-valve architectures are vulnerable for fatigue, leakage and even clogging from fluid impurities. It is for this reason that exploring controllable microfluidic vessels, may require extensively more research and development.

1.3 Fluid Pressure Control Platforms

Despite significant progress in closed-loop control theory, the number of studies applying low-cost microcontroller-based PID control to pressure-driven fluidic systems remains limited. Most existing microcontroller based implementations target either flowrate regulation or semi-steady pressure control, with response times measured in seconds rather than milliseconds. In contrast to Computer based platforms which provide much higher response times and wave resolution at a millisecond scale. This thesis aims to bridge the current gap in performance from computer based system to a microcontroller scale

A leading example of high-fidelity pressure waveform reproduction is presented by Vijayaratnam et al. (2021), whose work directly precedes and motivates the present thesis. Their study developed an advanced in vitro system designed to investigate the haemodynamic consequences of myocardial bridging, a condition in which a segment of a coronary artery becomes compressed during systole due to being embedded within myocardial tissue. To replicate this behaviour, the authors constructed a detailed bench-top model combining controlled mechanical compression, compliant arterial geometry, and patient-specific pressure tracking.

The experimental platform developed by Vijayaratnam et al. (2021)(Figure 6) incorporated several high-precision components arranged in a tightly integrated configuration. The core actuator was a laboratory-grade piston pump capable of delivering rapid, high-bandwidth pressure changes and reproducing patient-derived pressure waveforms with fine temporal resolution. Pressure was measured using a clinical-grade PressureWire catheter, which provided dual-site, high-fidelity recordings at the proximal and distal ends of the coronary model. The vessel analogue consisted of a scaled PDMS coronary artery with tuneable compliance, mounted inside a custom mechanical assembly that imposed rhythmic myocardial compression aligned with the cardiac cycle. All sensing, control, and actuation commands were orchestrated through MATLAB (MathWorks Inc.), which implemented an iterative PID-based control loop to refine the piston pump's input waveform after each cycle. This enabled the system to progressively minimise the discrepancy between the measured upstream pressure and the patient-derived target waveform. Together, these components formed an in vitro coronary simulator capable of reproducing physiologically accurate pressure environments.

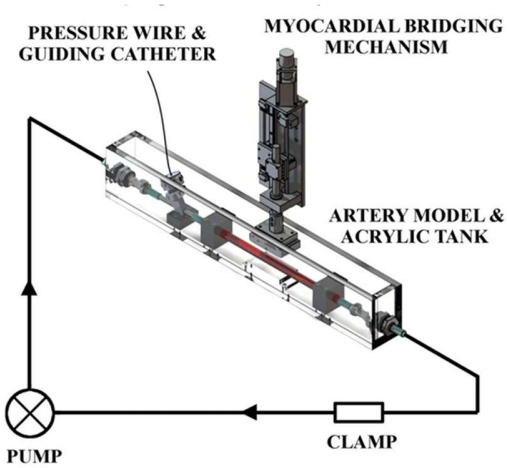


Figure 6 Vijayaratnam et al. (2021) Experimental Setup

Despite the system's complexity, the authors demonstrated that this configuration enabled exceptionally accurate reproduction of patient-specific aortic pressure waveforms. Using their iterative PID-based correction approach, the upstream pressure converged rapidly toward the target waveform, with mean absolute error decreasing from approximately 3 mmHg to 0.26 mmHg within just 12 iterations(Figure 7). The platform was able to reproduce key haemodynamic features, including the systolic upstroke, dicrotic notch, and diastolic decay, while also capturing the pressure distortions caused by dynamic myocardial compression.

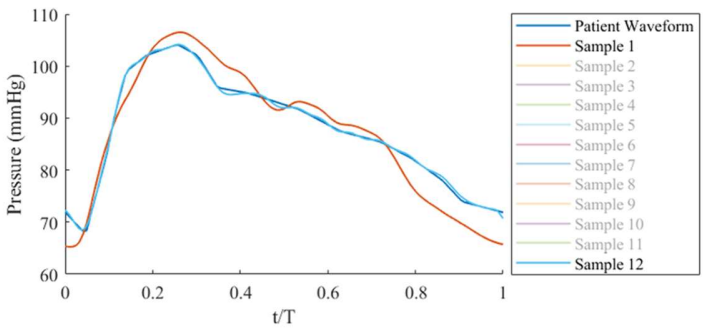


Figure 7 Patient Pressure waveform iterative results

Although extremely capable, this setup came with constraints in terms of cost, complexity, reproducibility limitations outside specialised laboratory equipment and real-time waveform fidelity. This project therefore follows on from Vijayaratnam's work by targeting the core functionality that PID-driven pressure control offers, as viable method for patient-specific waveform reproduction and seeks to adapt it to a low cost, Arduino-based syringe pump platform.

As an alternative, the pressure-waveform reproduction method demonstrated by Helms et al. (2021), who developed a modular hemodynamic simulator (MHS) capable of reproducing physiological and pathological waveforms through mechanical actuation and adjustable hydraulic elements. In contrast to Vijayaratnam et al.'s digitally controlled, PID-regulated piston pump, the MHS achieves waveform shaping entirely through tuneable physical parameters, namely resistance, compliance, valve competency, and pump configuration, without the use of computer-driven feedback. This mechanically centred design offers a valuable reference point for understanding how vascular geometry, compliance, and nonlinear resistance interact to determine haemodynamic behaviour in benchtop systems, while also highlighting the limitations of open-loop control in achieving precise or automated pressure tracking.

The MHS consists of three interchangeable modules (Figure 8): a pump module, a compliance chamber, and a bioreactor chamber. Flow is generated using modified peristaltic and roller pumps, with different pump configurations selected according to the target vessel type (e.g triple-roller pump for venous waveforms, roller pump from a heart–lung machine for high-flow aortic profiles). By removing or adjusting pump rollers, the authors engineered pulsatile outputs that mimics unique pressure characteristics in vitro. The upstream and downstream check valves emulated physiological resistance within the vessel, allowing the creation of conditions such as stenosis or insufficiency. The complain chamber, arranged as a side branch reservoir, provided a controllable Windkessel effect to the system. Varying chamber air volume, fluid levels, or clamp resistance adjusted for pulse pressure during systolic (upstroke phases) rise times and diastolic decay.

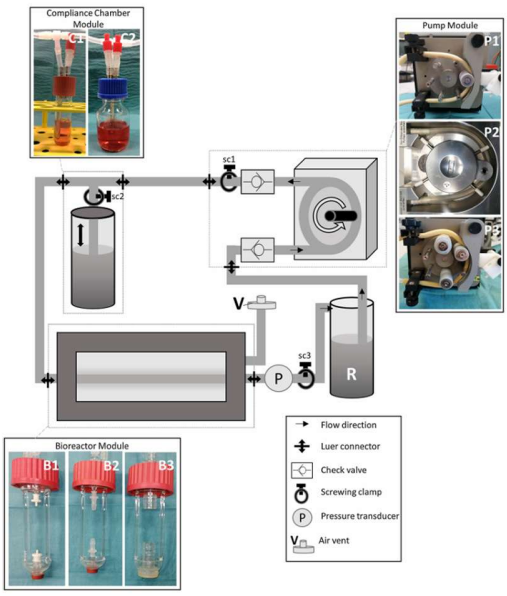


Figure 8 Helms et al. (2021) Experiment Setup

Using this modular architecture, Helms et al. successfully reproduced site-specific arterial and venous pressure waveforms, including characteristic features such as the aortic incisura and steep systolic upstroke of distal arteries as seen in figure 9.

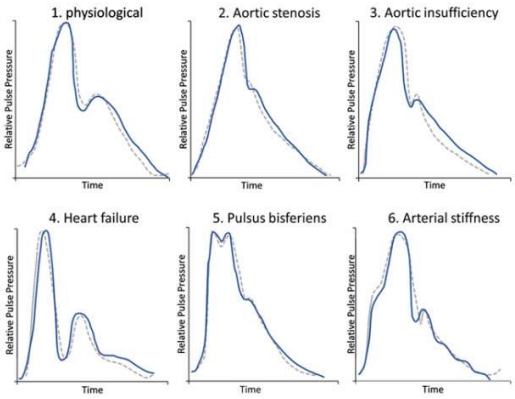


Figure 9 Helms et al (2021) Results

Despite its versatility, Helms et al.'s system exposes several limitations that are instructive for the development of digitally controlled, low-cost pressure platforms. Because the simulator relies entirely on mechanical actuation, each waveform must be produced through a unique combination of pump selection, compliance chamber adjustment, and resistive tuning, leaving no capacity for automated or programmable control. The use of a single mechanical pump inherently couples pressure and flow, preventing independent regulation and amplifying nonlinearities when distal clamps or resistive elements are altered. This manual tuning approach reduces reproducibility, limits temporal precision, and prevents closed-loop correction of waveform deviations. Nevertheless, the high accuracy demonstrated by the MHS highlights the performance that can be achieved when the mechanical components are carefully configured, reinforcing the importance of thoughtful equipment selection even when pursuing simpler, digitally controlled alternatives.

The first example of low-cost pressure regulation relevant to this thesis is presented by Lake et al. (2017), who developed an open-source syringe pump capable of maintaining stable inlet pressures in microfluidic channels using an Arduino-based PID controller. In contrast to higher-end platforms such as those of Vijayaratnam et al. (2021), which relied on MATLAB (MathWorks Inc.) and a laboratory piston pump, Lake et al.'s device demonstrates how effective pressure control can be achieved using inexpensive hardware, simple mechanical components, and embedded microcontroller logic.

Their system consists of a 3D-printed syringe pump actuated by a stepper motor, an inline piezoresistive pressure sensor amplified through an instrumentation amplifier, and an Arduino Uno responsible for data acquisition and PID computation (Figure 10). The pressure signal is sampled at 5 Hz (200 ms intervals), processed by the PID loop, and translated into stepper-motor motion to increase or decrease pressure within the microfluidic chip. This architecture differs fundamentally from both high-end iterative control methods

and mechanically tuned simulators: rather than relying on repeated waveform training or physical adjustment of compliance and resistance, Lake et al. achieve direct pressure feedback through low-cost sensors and real-time digital control.

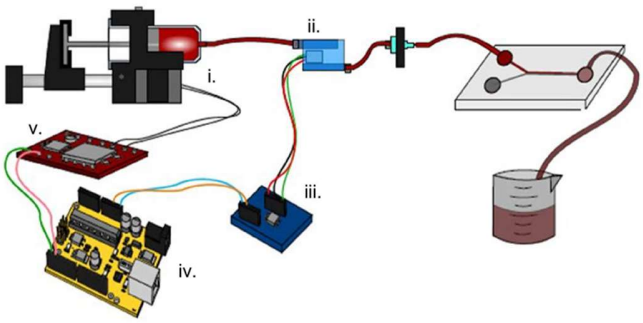


Figure 10 Lake et al.(2017) Experimental Setup

Using this configuration, the authors demonstrated that the Arduino PID controller could hold pressures within ±1% of the setpoint and achieve a settling time of approximately 1 s for small step inputs (Figure 11). They further showed that two pumps running independent PID loops could maintain equal pressures at a Y-junction inlet, eliminating the interface drift observed when commercial syringe pumps were operated in open loop. These results illustrate the core strength of microcontroller-based pressure regulation: even with inexpensive hardware and low sampling rates, closed-loop control substantially improves stability and repeatability compared with open-loop systems.

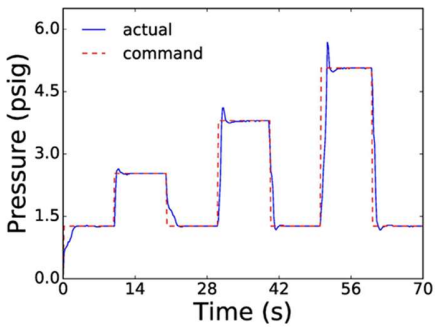


Figure 11 Lake et al. (2017) Results

However, the study also highlights several limitations that are critical when considering physiological pressure waveform reproduction. First, the sampling rate (5 Hz) severely restricts the controller's bandwidth, preventing accurate tracking of pulsatile signals that vary on millisecond or sub-second timescales. Second, the stepper-motor actuation provides only modest flow acceleration and cannot reproduce fast upstroke dynamics characteristic of arterial waveforms.

Lake et al.'s work is particularly informative, as it illustrates both the capabilities and the inherent limitations of a low-cost, microcontroller-based pressure platform. While the system demonstrates that meaningful pressure regulation can be achieved using inexpensive hardware and simple PID control, its restricted bandwidth, slow actuator dynamics, and sensitivity to downstream resistance highlight the challenges of reproducing physiologically realistic waveforms at low cost. Nonetheless, the study provides a valuable foundation for further development: it establishes the feasibility of closed-loop pressure control and offers experimentally validated methods and components that can be adapted or enhanced in subsequent designs

A more advanced example of digitally regulated pressure control is provided by Watson and Senyo (2019), who developed an all-in-one pneumatic microfluidics controller integrating onboard pressure generation for gas media and real-time PID regulation (Figure 12). In contrast to Lake et al. (2017), whose Arduino-based syringe pump provided low-bandwidth pressure regulation using a single sensor and stepper motor, Watson and Senyo constructed a modular, multi-channel architecture built around an ESP32 microcontroller, proportional solenoid valves, and custom-built embedded pressure regulators. Each pressure regulator contained its own ATmega32u4 microcontroller running a closed-loop PID algorithm at 1 kHz, allowing rapid adjustments through dual proportional valves that either increased pressure (inlet valve) or vented to atmosphere (outlet valve). Pressure sensing and control were fully embedded, with setpoints delivered via analogue pins, USB, or I^2C, enabling multiple regulators to be routed together. The system incorporated a small diaphragm pump, 12 V solenoid valves (actuated via ULN2803 drives), and a touchscreen interface, allowing manual and automated operation without additional laboratory infrastructure. This digital, multi-channel pneumatic platform demonstrates a level of integration far beyond previous low-cost designs, effectively bringing laboratory-style pressure control into a portable and open-source format.

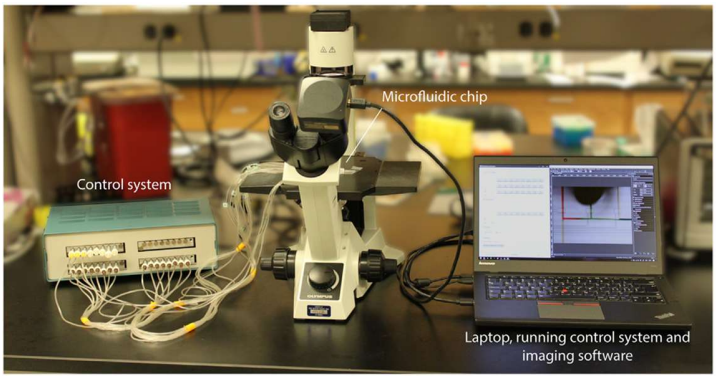


Figure 12 Watson and Senyo (2019) Experiment Setup

Performance testing showed that the embedded PID regulators achieved step responses time of 26-60 ms (depending on target pressure)(Figure 13) with steady-state errors below 3% across the 3–29.5 psi range. These dynamics were measured with 3.175mm tubing and minimal downstream flow, reflecting typical microfluidic dead volumes rather than high-flow vascular loads. While not designed for pulsatile physiological pressure replication, the system achieved stable and precise static pressure control suitable for a low-cost microcontroller based platform.

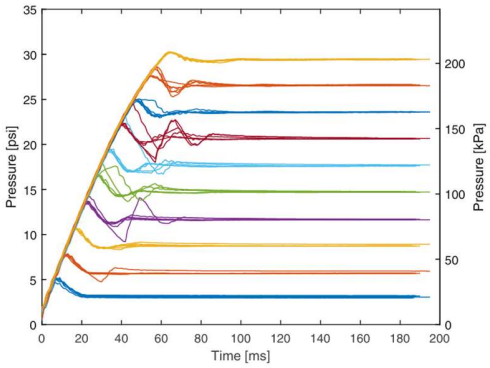


Figure 13Watson and Senyo (2019) Results

Despite its strong performance, Watson and Senyo's design also highlights several limitations relevant to low-budget physiological-pressure platforms. Firstly, the hardware complexity is substantial: custom PCBs, machined aluminium manifolds, proportional solenoid valves, and multi-board microcontroller coordination raise the cost (~USD $1730). Secondly, because the system regulates air pressure rather than hydraulic pressure, the measured performance does not directly translate to liquid-filled cardiovascular test loops, where compressibility, inertance, and tubing compliance introduce significant additional dynamics. Nonetheless, the system provides a valuable demonstration of how embedded PID, digitally addressed

regulators, and modular pneumatic components can be integrated into a compact, open-source control platform.

1.4 Arduino Based Sensor Adaptation

With a clearer understanding of the mechanical principles that shape physiological pressure waveforms and the limitations of existing pressure-mimicking devices, the final requirement for achieving accurate closed-loop control is the selection of a reliable feedback system and its integration method for microcontroller based platforms.

An example of low-cost Arduino-based sensor adaptation is described by Di Domenico (2020), who developed a multipurpose pressure transducer based device. The project aimed to replace expensive clinical monitoring hardware with a portable instrument capable of measuring intraluminal pressures in the 0-40 mmHg range using readily available components. The device combines an Arduino UNO (ATmega328P) microcontroller, a recycled Edwards TruWave disposable pressure transducer, a custom single-supply instrumentation amplifier based on the OP90 IC, a 16×2 I^2C LCD, a 9 V battery supply, and a simple plastic junction box enclosure. Pressure is acquired via the amplified bridge output, averaged over 1000 samples, and displayed in mmHg, with a push-button zeroing routine and optional serial data logging to a PC using CoolTerm. Calibration was performed against a water column from 0–35 cmH^2O, yielding a highly linear relationship between ADC counts and reference pressure (R^2≈1.00) and a regression equation used for real-time conversion (Figure 14). Bland–Altman analysis showed good agreement between the Arduino device and both the water column and a standard ICU monitor, despite the author noting that the device does not meet the formal accuracy standards of commercial clinical equipment. Overall, the study demonstrates that medical-grade pressure transducers can be successfully repurposed with Arduino and simple analog front-end circuitry to create robust, low-cost pressure measurement systems for laboratory research.

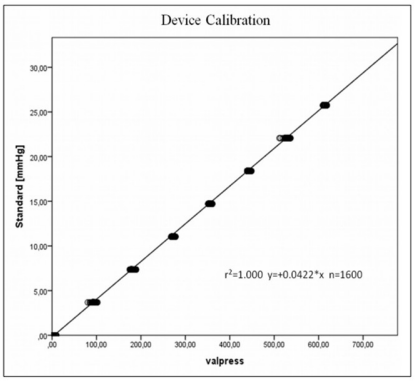


Figure 14 Di Domenico (2020), Sensor Calibration Results

1.5 Literature Review Conclusion

Across all studies, a common trend emerges, achieving high waveform accuracy typically requires significant investment in either hardware complexity or computational infrastructure. Conversely, lower-cost setups often sacrifice resolution, real-time adaptability, or feedback control. The gap between these two extremes defines the design space of this thesis to build a feedback-controlled, programmable, Arduino-based syringe pump capable of reproducing physiological pressure waveforms in a compact and cost-effective way, without the compromise of biological relevance or control accuracy.

2 Methodology

To replicate physiological pressure waveforms in real time, the system required several key components, defining the working parameters of the system. These included a liquid-media dispensing device capable of producing controllable flow (syringe pump), a pressure sensing unit to provide high-resolution feedback, a control platform for executing the PID algorithm and managing serial communication, and a fluidic circuit that could generate and sustain realistic pressure responses through adjustable hydraulic resistance. Together, these elements formed the essential subsystems of the final low-cost, real-time PID pressure waveform replication platform as seen below in figure 15.

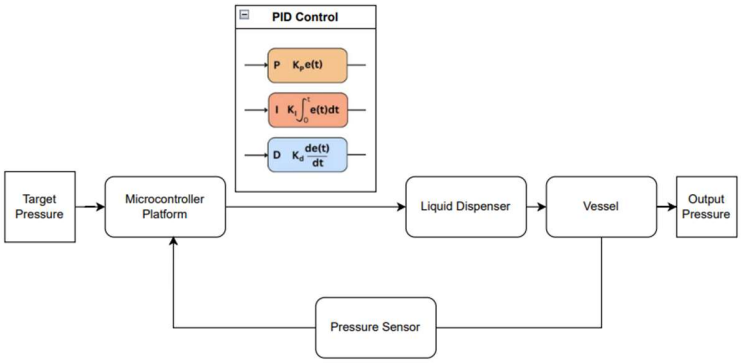


Figure 15 In Vitro PID Control Architecture

The development of the system followed an iterative approach aimed at progressively achieving the final integrated configuration. The process began with verifying the performance and compatibility of individual subsystems; including the liquid dispensing unit, pressure transducer, amplifier, and control hardware, to establish operational limits and ensure reliable communication between components. Once these elements were successfully integrated and calibrated, attention shifted to software adaptation, where PID control was implemented and refined to achieve stable closed-loop operation. Following this, the system was evaluated using reference physiological pressure waveforms to validate its ability to reproduce realistic patient-specific profiles.

2.1 Pressure Sensor Calibration and Verification.

To Achieve accurate and stable pressure feedback the Utah Medical Deltran disposable pressure transducer, a medical-grade analog sensor configured as a full Wheatstone bridge, was implemented. The device operates over a range of −50 to +300 mmHg with a nominal sensitivity of 5 µV/V/mmHg ± 2 %, providing the precision necessary for physiological pressure measurement. At a 5 V excitation voltage, the

transducer generates a full-scale differential output of approximately 7.5 mV, which, while highly linear, is below the resolution threshold of the Arduino's onboard 10-bit ADC.

The selection of a clinical-grade sensor follows precedents established by Vijayaratnam et al. (2021) and Helms et al. (2021), who emphasised the need for precise, low-noise feedback when reproducing physiological pressure waveforms. Moreover, Di Domenico (2020) demonstrated that re-purposed disposable medical transducers could be effectively integrated with open-source microcontrollers to deliver reliable measurements once properly amplified and calibrated.

2.1.1 Signal Conditioning

To interface the micro voltage of the Wheatstone bridge output with the Arduino based Controller, a HX711 24-bit digital bridge amplifier module was employed. Originally designed for load-cell applications, the module provides fixed programmable gains (up to 128) and differential input capability, converting the signal to a high-resolution digital stream accessible via a serial interface. This choice mirrors the instrumentation-amplifier methodology reported by Di Domenico (2020), who used an OP90-based circuit to achieve millivolt-scale amplification from a similar transducer. In both designs, low-cost amplification proved sufficient to capture clinically relevant pressures while maintaining compatibility with Arduino-level electronics.

The HX711 incorporates both an analog integrator and a digital decimation filter that reduce high-frequency noise through signal averaging, providing a default output rate of approximately 10 Hz. While this sampling rate is ideal for steady-state pressure, as seen in scale systems, it lacks the responsiveness required for real-time control applications. To address this, the module offers a high-speed operating mode. To enable a faster sampling rate, the module's RATE pin (Pin 15)(Figure 16.a) was switched from its default low-speed configuration (connected to ground) to the high-speed mode (connected to VCC). This was achieved by bridging the capacitor pad marked on the board to VCC, as shown in Figure 16.b, effectively increasing the sampling rate to around 80 Hz.

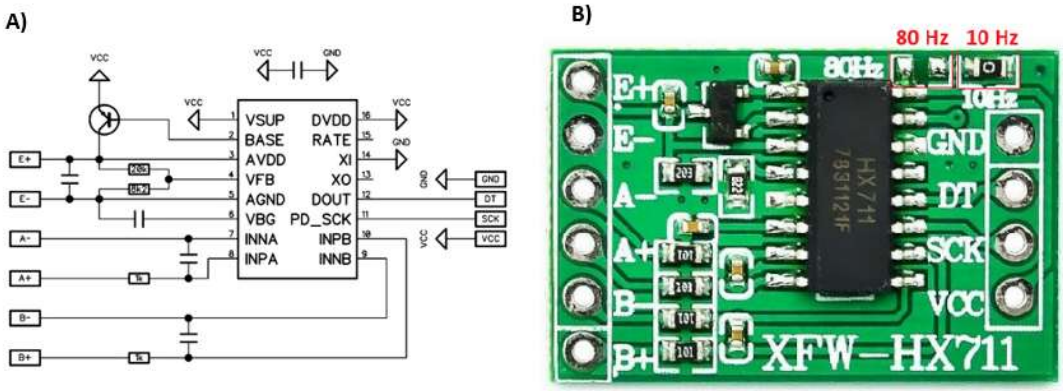


Figure 16 (A) HX711 Circuit diagram (B) HX711 ADC Amplifier - physical model displaying 10 and 80 Hz routing pads

2.1.2 Calibration Procedure

Following amplification of the Deltran transducer output, the system was recalibrated across its operating range using hydrostatic pressure laws via a static water column, where the pressure of the liquid is denoted by $P = \rho g h$. Calibration points were established at 60, 80, 100, 120, 140, 160, and 180 cmH^2O , representing physiological pressure range of approximately 44-132 mmHg. Prior to data collection, the sensor was zeroed at atmospheric pressure to account for any startup bias. The transducer was then connected to the base of the water column and carefully primed to eliminate trapped air (Figure 17.b).

At each height increment, a calibration script recorded the raw amplified digital output over a 5-second period, with ten repeated trials per pressure level. The resulting datasets were then averaged for run step, and the mean digital readings were plotted against the corresponding hydrostatic pressures (converted from cmH^2O to mmHg using a factor of 0.73556). Linear regression of these averaged values produced a calibration function relating analog-to-digital converter (ADC) counts to pressure in mmHg (Figure17.a). This calibration procedure followed the approach validated by Di Domenico et al. (2020), who demonstrated that hydrostatic column calibration combined with linear regression provides accurate and repeatable transducer characterisation for biomedical flow systems. As reported in their study, this method consistently achieved linear fits with coefficients of determination (R^2) exceeding 0.99,

confirming that the relationship between output signal and hydrostatic pressure can be treated as linear within the measurement range. The same procedure was therefore adopted to establish the scale factor used for all subsequent closed-loop pressure measurements.

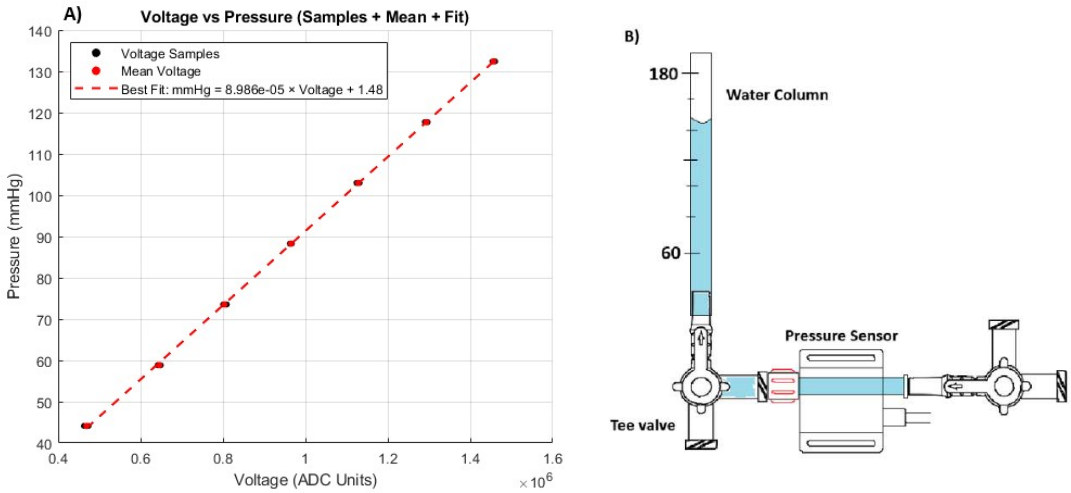


Figure 17 (A) Utah Medical Deltran Transducer Sample calibration data (B) Deltran Transducer Water Column Calibration Setup

2.1.3 Sensor Noise, Drift, and Filtering

To evaluate signal quality and stability under controlled conditions, the Deltran pressure transducer was interfaced with the Arduino Mega for data collection and tested under static pressures using the calibrated water column setup (Figure 17.b). Measurements were performed at 0, 100, 140, and 180 cmH^2O for 20 seconds at ambient room temperature. The objective of these tests was to characterise baseline noise levels, short-term drift, and sampling consistency prior to integration into the closed-loop control system.

Initial tests were conducted without any digital signal processing to capture the raw, unfiltered output from the HX711 amplifier. This established a baseline for evaluating the effectiveness of subsequent software-based filtering.

A digital moving-average low-pass filter was then implemented to attenuate high-frequency noise components and improve pressure signal stability. The filtered output at each sampling step, Y_{output}, was computed according to:

$$Y_{(output)} = \frac{1}{m}\sum_{i=0}^{m-1} Data_{n-i} \tag{4}$$

This equation computes the filtered output $Y_{(output)}$ by averaging the last m samples of the data stream, the averaging window length m was selected empirically to balance noise fluctuations without affective real-time resolution of data.

Each filtered dataset was subsequently analysed to quantify the reduction in noise variance and drift over time. These analyses enabled determination of suitable filter parameters for real-time feedback control.

2.2 Liquid Media Dispensing System

The liquid dispensing subsystem employed a New Era NE-1010 programmable syringe pump (Research Resource) as the sole actuation source for generating flow within the hydraulic circuit. The NE-1010 operates as a flow-rate-controlled device, driven by a microcontroller-based stepper motor and precision lead-screw assembly that advances the syringe plunger at a digitally defined rate. The pump supports programmable infusion sequences across a range of 1.459 μL/hr to 127.2 mL/min, depending on syringe size, and is capable of operating in both infusion and withdrawal modes.

Communication with the syringe pump can be established through RS-232 serial communication via its ethernet to DB9 interface, configured to a factory baud rate of 19200 bps. To enable compatibility with the Arduino Mega 2560, an RS-232-to-TTL(Figure 18.c) based on the SP2323 integrated circuit was utilized. This module converts the Arduino's native 5 V TTL logic levels to the $\pm$12 V RS-232 voltage levels required by the NE-1010, allowing both systems to connect via DB9 ports (Figure 18.a). However, because both the NE-1010 and Arduino are classified as Data Terminal Equipment (DTE), their communication protocols require a unique connection type. Unlike standard DB9 connection, where Receive (RX) and Transmit (TX) pins are connected directly in a TX-TX and RX-RX configuration, Two DTE devices require a null-modem (crossover) configuration which connects RX-TX and TX-RX as seen Figure 18.b

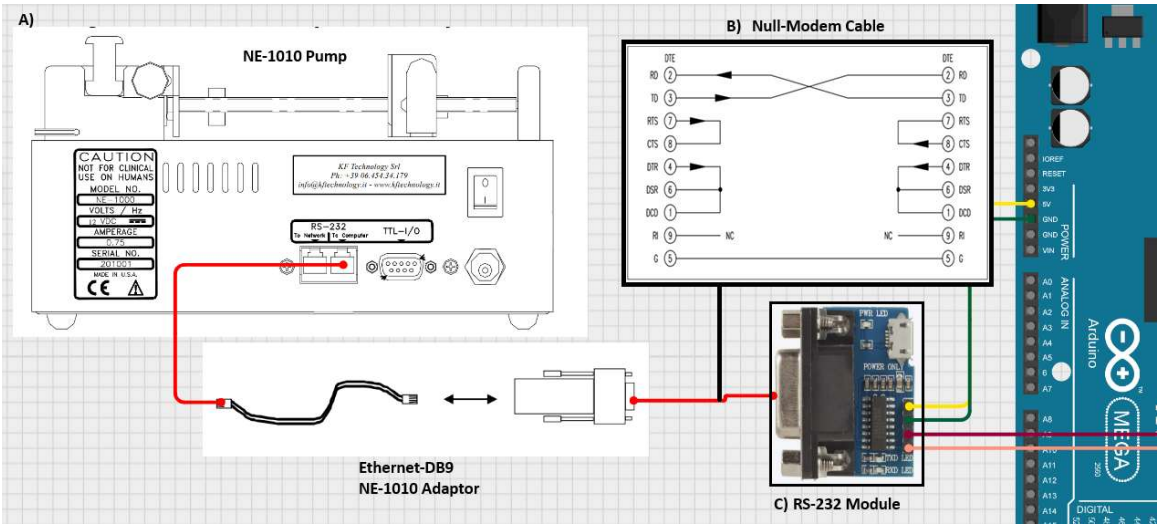


Figure 18 (A) Arduino Mega Connection Setup With NE-1010 Syringe Pump (B) Null Modem Cable Configuration (C)RS-232 Arduino Module

Connection can then be verified through Arduino Serial 1 comm, where all command and queries can be transmitted in the pumps' Basic Command Mode, following the syntax:

<address> <command> [parameter] <CR>

Allowing for parameter configurations such as syringe diameter (00 DIA <size>), Rate (00 RAT <value> <units>), as well as start (00 RUN) and stop (00 STP) commands. Apon receiving a correct command the NE-1010 will transmit a carriage return (CR) character, or an error, thus confirming communication.

2.3 Hydraulic resistance and laminar flow

The downstream vessel assembly was designed to generate controllable hydraulic resistance while maintaining laminar flow conditions in accordance with Poiseuille's law. The fluid pathway was constructed using Luer-lock connections for all junctions to ensure a sealed and modular setup. Flow from the syringe pump passed through a T-junction, with exit (1) connected to the Deltran pressure transducer and exit (2) serving as the outlet. This outlet connection was used to test multiple configurations that modified the system's effective hydraulic resistance and flow behaviour (Figure 19). It is important to note that all procedures are conducted using H^2O as the liquid media.

Three outlet types were evaluated:

A. 0.6 mm ID × 30 mm stainless-steel needle – representing a rigid, high-resistance laminar channel.
B. Adjustable valve outlet – provided variable flow restriction for coarse resistance tuning.
C. 0.5 mm ID × 15 mm needle coupled with an additional 25 mm × 0.5 mm ID semi-rigid capillary tube, providing an extended fixed-resistance path with negligible compliance.

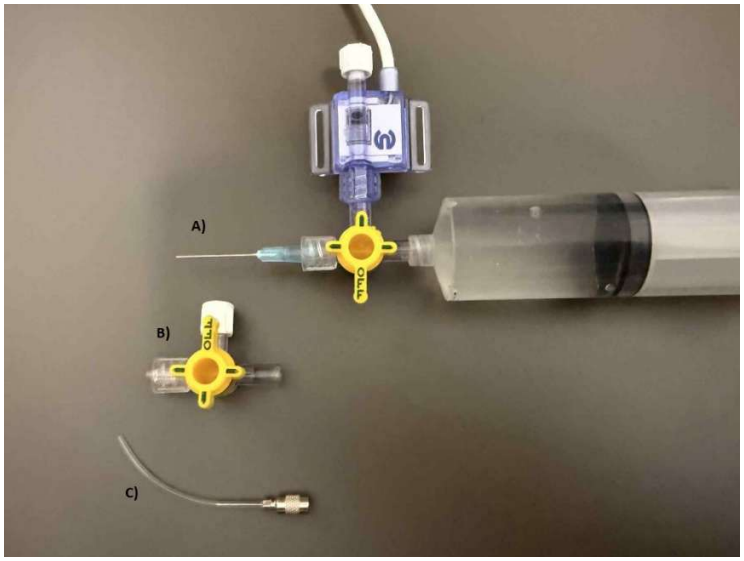


Figure 19 In Vitro Outlet Configuration - (A) 0.6 mm ID needle, (B) Tee adjustable valve, (C) 0.5 mm ID Tubing + Needle

The testing approach was modelled on the syringe-based resistance characterisation method using Reynolds-number and linear regression models described by Patil et al. (2014) and guided by the analytical framework presented by Lee et al. (2022), which applied Bernoulli and Poiseuille principles to relate outlet diameter and flow velocity to pressure development in syringe systems.

To confirm that the experimental flow conditions remained within the laminar regime assumed by Poiseuille's law, the Reynolds number (Re) was estimated for each outlet configuration (Table 1) using:

$$Re = \frac{\rho v D}{\mu} = \frac{4\rho Q}{\pi D \mu} \quad (5)$$

Reynolds number Re indicates laminar flow for values below 2000, where Re is a product of fluid density ρ = 998 kg m^{-3}, Volumetric flowrate Q, dynamic viscosity μ =1 $\times 10^{-3}\ Pa \cdot s\ at\ 20°C$, and vessel diameter D.

Table 1Reynold's Number Outlet Analysis

Flowrate mL/min	Needle 0.6 (ID) x 30 mm	Needle + Tubing 0.5 (ID) x 40 mm	Valve NA
4	141.19	169.43	NA
8	282.38	338.85	NA
12	423.56	508.28	NA
16	564.75	677.70	NA
20	705.94	847.13	NA

Because the syringe outlet and connector cross-sections had significantly larger internal diameters than the outlet, the dominant viscous losses were assumed to occur across the needle or capillary segment. Therefore, Reynolds numbers were calculated using only the outlet geometry, following the approach established by Patil et al. (2014) and Lee et al. (2022). Both studies demonstrated that for syringe-based flow systems, the pressure drop across upstream fittings and taps is negligible compared with that across the needle bore, and that laminar flow can be verified by a linear ΔP–Q relationship.

Following these established methodologies, steady-flow tests were conducted at flowrate setpoints of 4, 8, 12, 16, and 20 mL min^{-1}, covering the operational mid-range of the NE-1010's linear actuation system and health physiological pressure ranges. Each outlet configuration was tested five times per flowrate to ensure repeatability. The system was allowed to reach steady-state equilibrium before acquisition, and mean pressure values were recorded using the calibrated Deltran transducer. All data were streamed from the Arduino Mega via serial interface and logged as CSV files using the Arduino IDE Serial Monitor. The recorded datasets were imported into MATLAB (MathWorks Inc.) for averaging and least-squares linear regression fitting of the pressure–flow relationship:

$$P = RQ \tag{6}$$

Where, pressure P is proportional to the flowrate Q, under some resistive condition R.

Following the linearity assessment, the selected outlet was further evaluated for its passive pressure-decay behaviour. This experiment was performed by first establishing a stable steady-state pressure using flow-rate inputs of 12, 14, and 20 mL/min, after which the pump was abruptly stopped. Pressure data were recorded for one second prior to pump termination to capture the pre-decay behaviour, and continuously thereafter to observe the natural pressure fall within the vessel. The resulting pressure traces were imported into MATLAB (MathWorks Inc.) for analysis, where the decay response was examined at the inflection

point marking the transition time from 10 mmHg prior and after. This analysis allowed the passive pressure-decay rate to be quantified and compared against diastolic pressure phases.

This testing procedure allowed each outlet configuration to be characterised in terms of hydraulic resistance and linearity, providing the basis for selecting the most predictable outlet for subsequent closed-loop control experiments.

2.4 Final System Configuration and Cost

After verification of all individual subsystems, the complete closed-loop pressure control platform was assembled for all subsequent PID control optimisation. The final configuration consisted of the NE-1010 syringe pump (flow actuation for H^2O liquid media), the Utah Medical Deltran transducer with HX711 amplifier (pressure feedback), and the Arduino Mega 2560 (digital controller and data acquisition unit). All communication between modules was handled through serial interfaces: RS-232 for pump control, I²C for pressure signal conditioning, and UART for real-time data logging (Figure 20 and 21.A).

The hydraulic circuit was configured using a 60 mL syringe connected through a T-junction to both the pressure transducer and outlet assembly (Figure 20.B). Based on the preceding resistance tests, the 0.6 mm ID × 30 mm stainless-steel needle was selected as the outlet, as it demonstrated a near-linear pressure–flow relationship while maintaining stable operation within the mechanical limits of the pump. A micro-SD module was integrated to store time-stamped pressure and large higher resolution control data for PID control

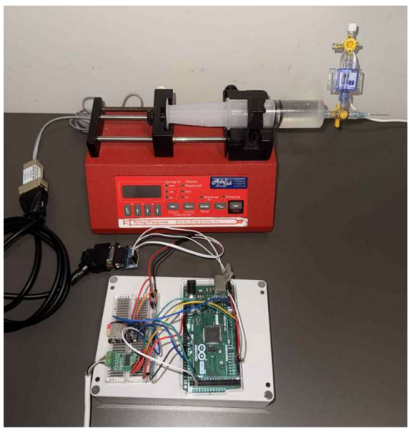


Figure 20 Final Experimental Setup

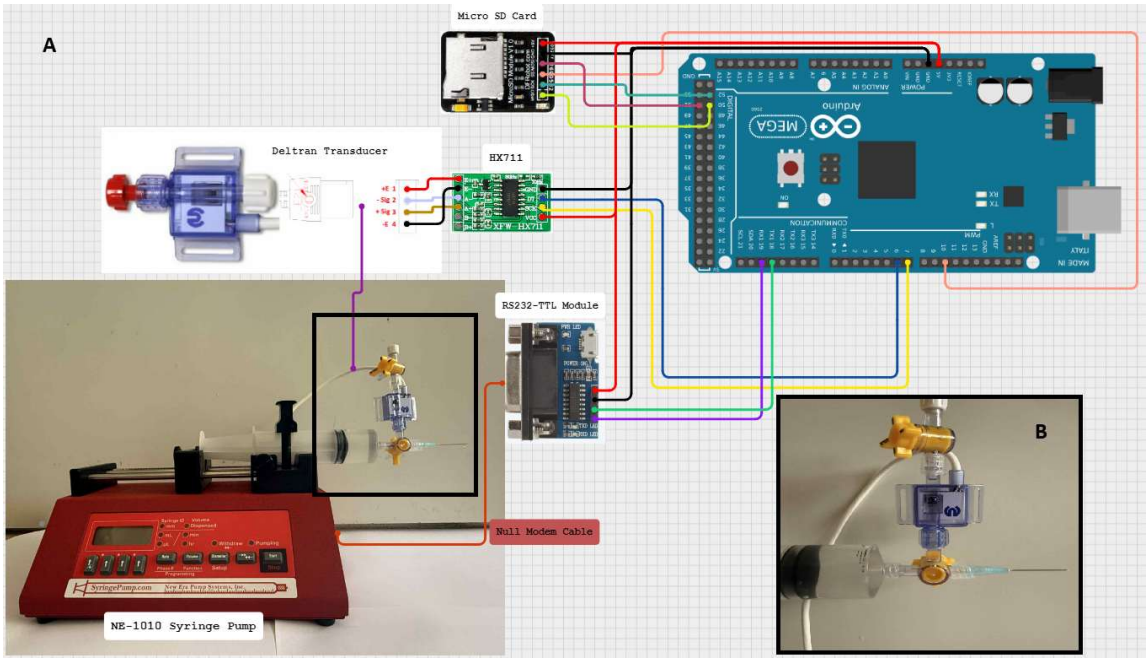


Figure 21 (A) Experimental Assembly (B) Hydraulic Vessel Configuration

To demonstrate the accessibility and low-cost nature of the platform, the approximate equipment expenses were summarised in Table 2. Excluding the NE-1010 pump, which was provided as a research resource, the total hardware cost of the system was approximately $140 AUD, underscoring the affordability of the proposed approach relative to commercial pressure-control systems.

Table 2 Equipment Cost - Final Setup

Equipment	Cost
Arduin Mega	$35 AUD
Pressure Transducer	$30 AUD
Miscellaneous (Tubing, Valves, etc)	$30 AUD
HX711 Module	$3 AUD
RS232-TTL Module	$30 AUD
Micro SD module	$12 AUD
New Era NE-1010	(Research resource)
Total	$140 AUD

2.5 Control Architecture

The control algorithm was implemented on the Arduino Mega 2560 using a discrete-time proportional–integral–derivative (PID) feedback loop, which continuously adjusted the commanded syringe-pump flow rate based on the instantaneous error between measured and target pressure. Microcontroller-based systems inherently operate in discrete time because sensor readings, computations, and actuator commands occur at finite sampling intervals. As highlighted by Rehman et al. (2020), continuous-time PID laws cannot be executed directly on digital processors and must first be expressed in discrete form.

Accordingly, the PID control law implemented on the Arduino was:

$$u[n] = K_p e[n] + K_I \sum e[i]\Delta t + K_D \frac{e[n] - e[n-1]}{\Delta t} \tag{7}$$

The control output u[n]was transmitted to the NE-1010 syringe pump via RS-232 at 19,200 bps, issuing real-time flow-rate commands in the pump's Basic Command Mode. Although the Deltran–HX711 acquisition chain is capable of a theoretical 80 Hz data rate, the combined overhead of sensor acquisition, PID computation, serial communication, and SD-card logging resulted in an effective control-loop frequency of ~77 Hz, which served as the practical discretisation interval for the controller.

2.5.1 PID Tuning Approach – Ziegler-Nichols Method

The controller was initially tuned using the open-loop Ziegler–Nichol's method (Ziegler and Nichols, 1942), serving as the first iteration of the PID control design. This classical method determines controller parameters from the process's open-loop step response, providing a structured framework for initial gain selection. Although more modern adaptive and optimization-based tuning strategies, such as Internal Model Control (IMC-PID) and Reinforcement Learning, exist, these approaches require detailed knowledge of the system's transfer function or dynamic model to define performance objectives and cost functions. In contrast, the empirical Ziegler–Nichol's approach relies solely on measured response data, making it more practical for low-cost, microcontroller-based pressure control systems where full system modelling is unachievable.

To characterise the system response following the open-loop method, a step input in flow rate was applied to the syringe pump while pressure feedback was recorded under open-loop conditions (no feedback

control). All pressure and flow data were streamed from the Arduino Mega via the serial interface and stored as CSV files for post-processing. The recorded datasets were imported into MATLAB (MathWorks Inc.) for numerical analysis and visualisation of the process reaction curve. In accordance with the Ziegler–Nichol's approach, the resulting pressure response was required to exhibit a characteristic S-shaped curve with a stable steady-state value. This behaviour was obtained empirically through incremental adjustments to the step amplitude until the pressure response reached the target physiological pressure range (HPR). From this curve, key process parameters were extracted: the process gain K, determined from the ratio of pressure rise to flow-rate step,

$$K = \frac{\Delta y}{\Delta u} = \frac{85}{13} = 6.54 \frac{mmHg}{\left(\frac{mL}{\min}\right)} \tag{8}$$

and the time constant T and dead time L, derived from the tangent drawn at the inflection point of the curve where the rate of pressure change was maximal (Figure 22)

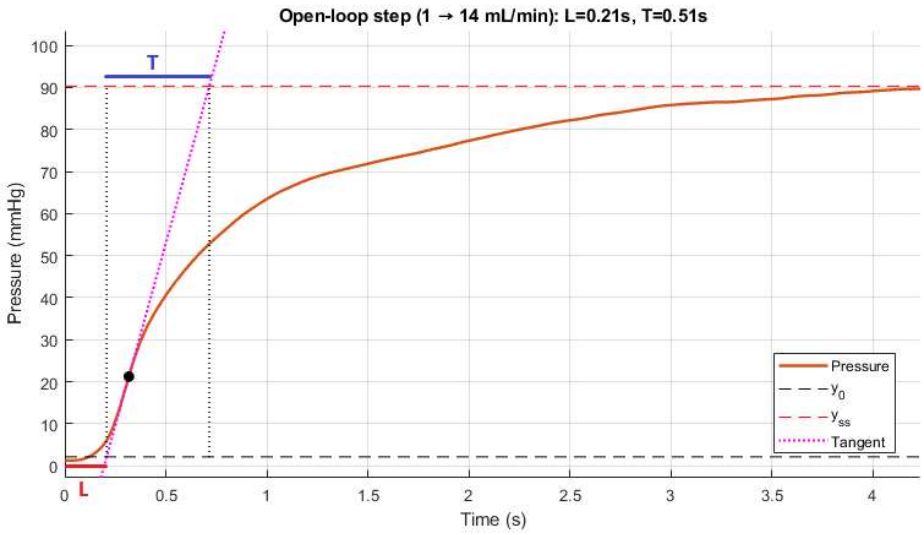


Figure 22 Ziegler-Nichols Experimental Response

With the process parameters K, L, and T identified from the open-loop step response, the initial controller gains were computed using the Ziegler–Nichols open-loop tuning rules (Table 3).

Table 3 Ziegler-Nichol's Controller Gain Parameters

Controller Type	K_p	T_I	T_D
P	$\frac{T}{KL}$	–	–
PI	$\frac{0.9T}{KL}$	$3.33L$	–
PID	$\frac{1.2T}{KL}$	$2L$	$0.5L$

Using the experimentally obtained process parameters:

$$K = 6.54, \qquad L = 0.21s, \qquad T = 0.51s \tag{9}$$

The continuous-time Ziegler-Nichols PID gains were computed as:

$$K_p = 0.4456, \qquad T_I = 0.42s, \qquad T_D = 0.105s \tag{10}$$

For implementation on the Arduino Mega, the continuous-time PID parameters were converted to their discrete-time equivalents using the standard forward-Euler discretisation:

$$K_p^{discrete} = K_P = 0.4456 \tag{11}$$

$$K_i^{discrete} = \frac{K_p}{T_I} = 1.061 \tag{12}$$

$$K_d^{discrete} = K_p T_D = 0.0468 \tag{13}$$

2.5.2 Closed-Loop PID Evaluation Using Custom Input Pressure Waveforms

After completing the initial Ziegler–Nichols tuning and empirical refinement under step-response conditions, the controller was evaluated using more realistic, dynamically varying pressure trajectories. These tests were performed prior to physiological waveform tracking to validate the controller's ability to follow complex, time-varying references within the healthy physiological pressure range (HPR).

Controlled input pressure profiles were generated in MATLAB (MathWorks Inc.) using the discrete sampling interval derived from the measured control-loop timing (~77 Hz). This ensured that each waveform was discretised at the exact temporal resolution of the Arduino controller. The data is then uploaded to a micro SD card in CSV format for arduino to sample during real-time.

The four constructed input profiles were designed to represent a range of tracking difficulties, including slow linear ramps, abrupt step-like rises, and smooth curved transitions as seen in figure 23. Each waveform was executed as a continuous sequence of four repeated cycles, where every cycle began with a stabilised

baseline pressure of 80 mmHg followed by the waveform's pulse segment. Running four cycles per experimental run allowed the controller's repeatability, stability, and drift behaviour to be evaluated under identical conditions. The use of repeated cycles also provided an opportunity to refine the PID gains while observing how the controller responded to different waveform gradients and dynamic demands.

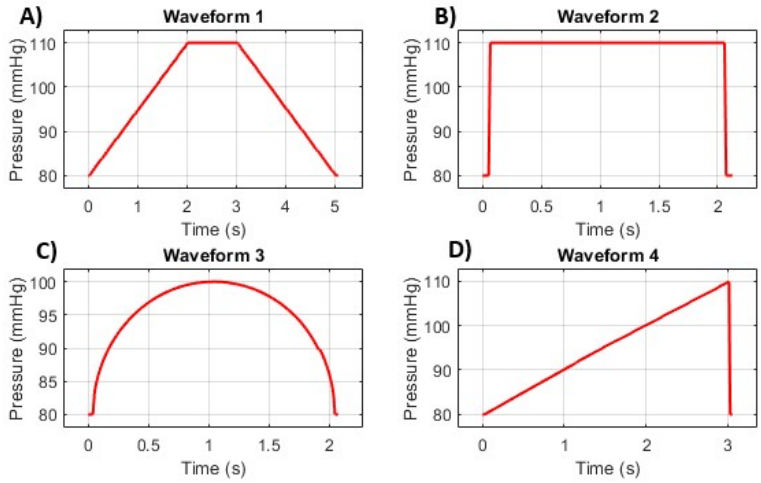


Figure 23 Custom Controlled Input Profiles - (A) Trapezoidal (B) Square (C) Circular (D) Triangular

2.6 Physiological Pressure Waveform Replication

With the PID gains optimised under synthetic waveform conditions, the controller was next evaluated using physiologically realistic pressure profiles. Three arterial pressure waveforms, seen in Figure 24 (Aortic root, Carotid, and Iliac), were sampled from *the virtual database of healthy subjects published* by Willemet et al. (2015). Although this database contains waveforms through several different arterial locations, the three specified were chosen due to their varying morphologies.

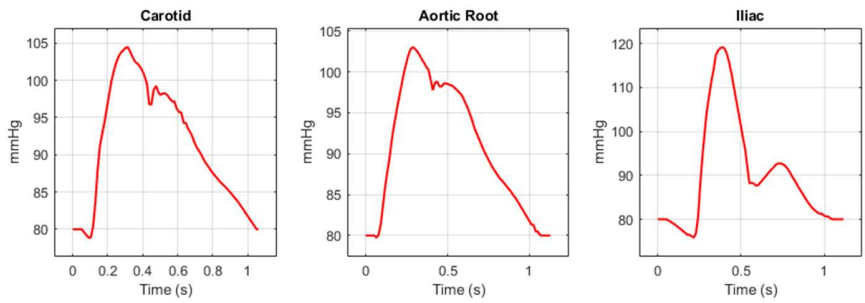


Figure 24 Original Physiological Waveforms (sourced from Willemet et al (2015)) – All data points are shift to a baseline pressure of 80 mmHg

The testing procedure followed the same structure as the synthetic waveform testing (Section 2.4.3) and final setup configuration (Figure 21.a). Each cycle began with a fixed baseline pressure segment, followed by a dynamic waveform segment. To investigate the limits of the controller's dynamic response, each arterial waveform was tested at three cardiac cycle lengths (wave frequencies): the original 1-second cycle and stretched 2-second and 4-second variants. Waveform adaptation was performed in MATLAB (MathWorks Inc.), where the original pressure values (provided in Pascals) were converted to mmHg, baseline-shifted to the experimental 80/120 mmHg range, and resampled at 77 Hz to match the measured control-loop timing of the Arduino Mega. The temporal stretching was achieved through linear interpolation of the original waveform onto the desired time base, ensuring that all derived signals preserved their physiological morphology while matching the discrete-time resolution of the controller.

To quantitatively assess pressure-tracking performance, the recorded waveforms were evaluated through MATLAB (MathWorks Inc.) using standard error metrics commonly employed in physiological waveform validation studies. For each waveform, the mean absolute error (MAE), root-mean-square error (RMSE), peak error, and normalised RMSE (NRMSE) were computed relative to the corresponding digital reference signal. This analytical procedure follows approaches used in prior experimental haemodynamic and pump-control studies, such as those by Vijayaratnam et al. (2021), Helms et al. (2021), and Lake et al. (2024), where these metrics are used to characterise waveform fidelity, amplitude accuracy, and cycle-to-cycle repeatability. The calculated values allow direct comparison across different waveform types, durations and provide a consistent framework for evaluating the accuracy and stability of the closed-loop control system.

3 Results and discussion

In this section, the developed methodology is evaluated to characterise the behaviour and limitations of the current system, as well as to assess its closed-loop, real-time performance under physiological pressure-tracking conditions.

3.1 Pressure sensor stability

Table 4 summarises the standard deviation and error-from-mean values obtained for both unfiltered and filtered signals recorded by the Utah Medical Deltran transistor. The unfiltered dataset exhibited a standard deviation of 0.11 mmHg and a maximum magnitude 0.45 mmHg, indicating that the raw signal is inherently stable even without post-processing. As observed in figure 25, the implemented moving-average filter resulted in a modest improvement, reducing the standard deviation to 0.036 mmHg and limiting deviation from the mean to 0.11 mmHg. While the reduction in noise is measurable, the overall improvement is minor relative to the already low baseline variability. This confirms that the sensor behaviour is fundamentally stable and that filtering, although beneficial, does not substantially alter system performance.

In addition to noise characterisation, this test confirms the practical sampling limits of the HX711 module, at approximately 80 Hz, measured by the number of samples collected in a 20s experiment (Figure 25). This constraint is critical, as it directly influences the maximum achievable control bandwidth and therefore the fidelity of real-time pressure control

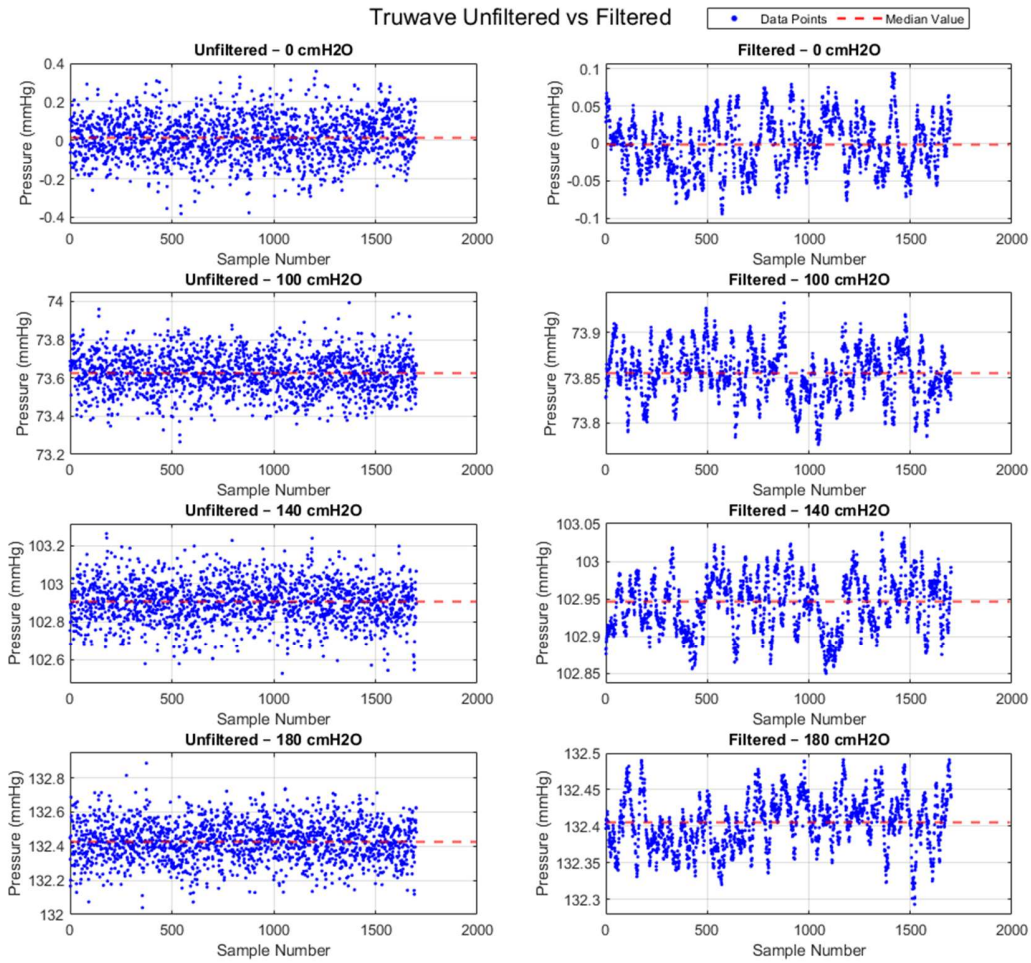


Figure 25 Utah Medical Deltran Static Pressure Data

Table 4Utah Medical Deltra Static Pressure Analysis (Mean value, STD mmHg, Max error between mean and measured, Min error)

PressureData	Mean_mmHg	Std_mmHg	ErrMax	ErrMin
Unfiltered 0	0.013724	0.11062	0.34428	0.39472
Unfiltered 100	73.624	0.1014	0.36889	0.35811
Unfiltered 140	102.9	0.1086	0.35862	0.37638
Unfiltered 180	132.43	0.11386	0.45695	0.38905
Filtered 0	-0.0016745	0.034856	0.095675	0.092325
Filtered 100	73.854	0.027975	0.078655	0.078345
Filtered 140	102.94	0.036509	0.094376	0.093624
Filtered 180	132.4	0.035735	0.08676	0.11124

3.2 Hydraulic Resistance Results

The hydraulic resistance of the tested outlet configurations was evaluated by measuring the steady-state pressure generated across a range of imposed flowrates. Across all outlets, the pressure–flow relationship exhibited a characteristic upward curvature, where pressure increased at an incrementally higher rate as flowrate rose. This behaviour aligns with reports in the literature describing non-linear pressure development in needle-based and small-diameter outlet geometries, where viscous losses rise disproportionately at higher velocities. Similar curvature has been documented by Lee et al. (2022), who observed an exponential-like increase in syringe outlet pressure for lower-gauge needles, with an asymptotic range at low flowrates before accelerating rapidly with increasing volumetric rate. The same qualitative pattern is apparent in Figure 26, confirming that the present system reproduces expected hydraulic behaviour for laminar flow through constricted outlets.

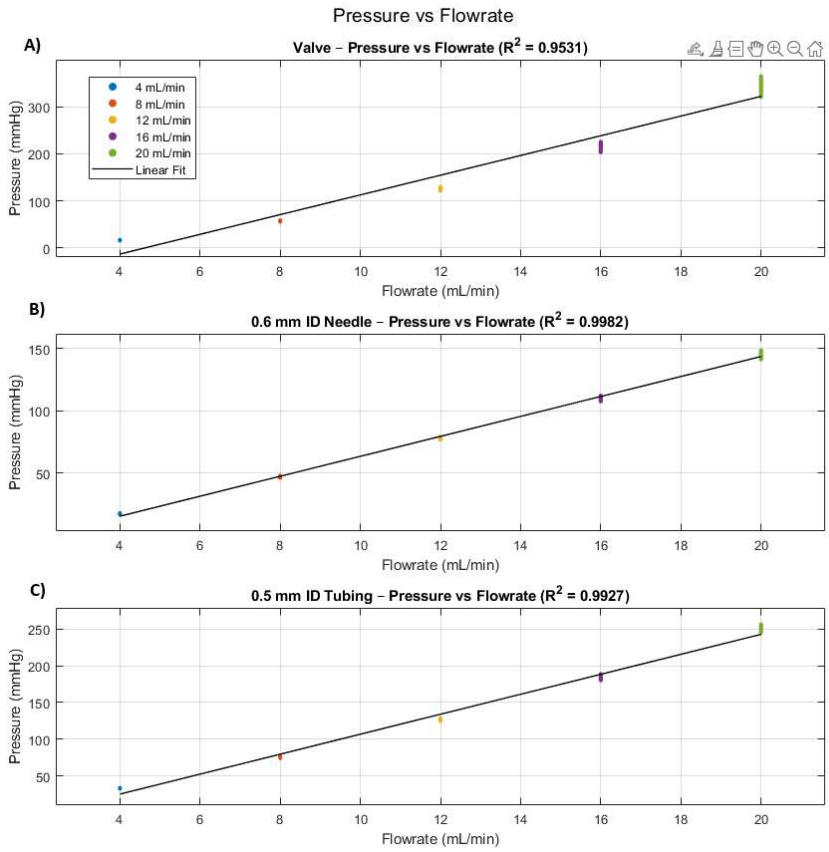


Figure 26 Pressure vs Flowrate measure at 8,12,16,20 mL/min for three outlet configurations - (A) Controllable Valve, (B) 0.6 mm ID Needle, (C) 0.5 mm ID Tubing + Needle

Among the tested outlets, the 0.6 mm inner-diameter needle produced the most linear and predictable response, with the smallest deviation from the fitted line as seen in table 5. The maximum observed pressure error remained within approximately 2 mmHg, making it the most favourable option for repeatable pressure modulation. Despite its improved consistency, the underlying curvature was still present, particularly at higher flowrates, with pressure data displaying a considerable amount of variation. This widening spread of pressure at elevated flowrates could not be attributed to a singular cause, however the most likely factor could be the drive mechanism explored in the following section.

Table 5 Outlet Configuration Results

Setup	Flow_mLmin	SampleMean_mmHg	ExpectedLinearFitValue_mmHg	AbsError_mmHg
Valve	4	16.24	-12.827	29.067
Valve	8	57.536	70.87	13.334
Valve	12	126.86	154.57	27.702
Valve	16	217.4	238.26	20.862
Valve	20	354.79	321.96	32.831
0.6 mm ID Needle	4	17.23	15.491	1.7388
0.6 mm ID Needle	8	46.831	47.493	0.66138
0.6 mm ID Needle	12	77.806	79.494	1.6885
0.6 mm ID Needle	16	109.9	111.5	1.5941
0.6 mm ID Needle	20	145.7	143.5	2.2051
0.5 mm ID Tubing	4	33.038	25.242	7.7966
0.5 mm ID Tubing	8	75.203	79.627	4.424
0.5 mm ID Tubing	12	127.02	134.01	6.9935
0.5 mm ID Tubing	16	184.47	188.4	3.9274
0.5 mm ID Tubing	20	250.33	242.78	7.5483

Based on the linearity assessment, the 0.6 mm inner-diameter needle. To further characterise its decay behaviour at 12, 14, and 20 mL/min. Pressure data was recorded from one second prior to flow termination to capture the behaviour of pressure within the vessel. The resulting pressure traces (Figure 27) indicate a slight carry-on of pressure for 0.2 seconds past termination, which is presumed to arise from residual potential energy in the syringe as it settles towards equilibrium with the remaining pressure in the system. The passive pressure decay rates were measured at 48.669, 67.416 , 88.64 mmHg/s, respectively, with slower decay at lower internal vessel pressure, confirming the vessel's capacity to dissipate at a rate similar to that observed in systolic decline.

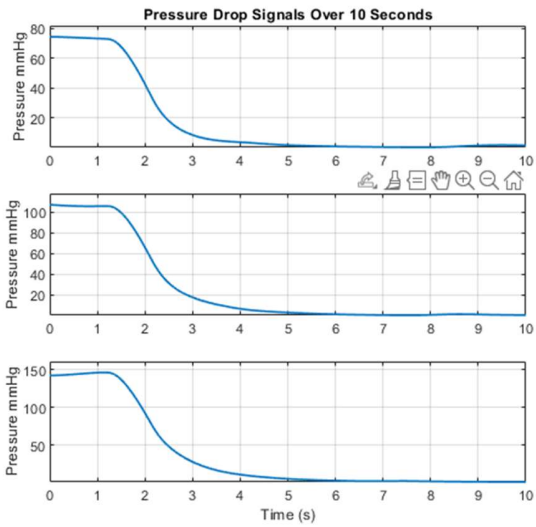


Figure 27 0.6 mm ID Needle Pressure Drop Results

3.1 Pump Oscillation Results

To investigate the source of the pressure variation observed at higher flowrates in Section 3.2, the behaviour of the syringe pump was analysed under steady, constant-flow conditions using an identical test configuration. The resulting pressure traces (Figure 28) confirm that both NE-1010 dispensing units exhibit a characteristic oscillatory pattern superimposed on the nominal steady-state pressure. These oscillations occurred at consistent frequencies for a given flowrate and increased proportionally to input (Table 6), indicating that the effect originates from the internal drive mechanism of the pump rather than from the hydraulic outlet or vessel configuration.

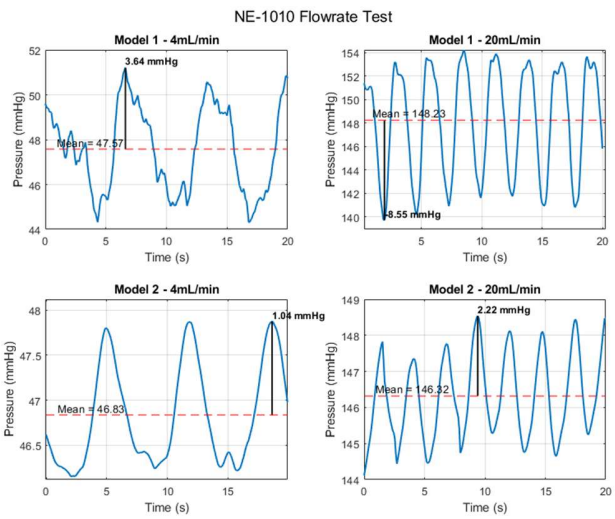


Figure 28 NE-1010 Comparison Results at 4, 20 mL/min, Old Model (1), New Model (2)

Comparison of the two pump models revealed that both devices produced the same fundamental frequency behaviour, but with differing amplitudes. Model 1 generated larger oscillation peaks, whereas Model 2 demonstrated moderately reduced oscillation magnitude, as illustrated in figure 5. Despite this improvement, both pumps still introduced a persistent periodic pressure ripple of approximately 1-2 mmHg, representing an inherent mechanical limitation of the pump's actuation system.

Table 6 Model 1 Vs Model 2 Frequency Results

	8 ml/min - Frequency	20 ml/min - Frequency
NE 1010 (model 1)	~ 0.149 Hz	~ 0.379 Hz
NE 1010 (model 2)	~ 0.139 Hz	~ 0.371 Hz

This oscillation imposes a lower bound on achievable pressure smoothness and contributes to the widening spread of measured pressure at higher flowrates noted in Section 3.2 and seen in figure 29. In practical terms, this behaviour constitutes a systematic mechanical limitation of the pump, and therefore a non-negligible disturbance that the closed-loop controller must compensate for in later sections.

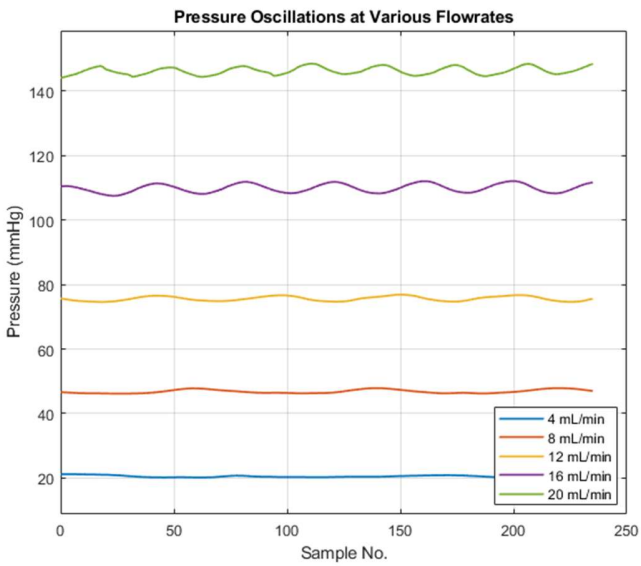


Figure 29 Pressure oscillation for improved pump and 0.6 ID mm Needle outlet configuration

3.3 Custom Input Waveform Reproduction Results

The Refined PID controller was first evaluated using the synthetic stepwise pressure trajectories outlined in section 2.3.4. This trajectory exposes system limitations which carry on from the previous sections. As seen in Figure X the system demonstrated substantially better fidelity when following gradual, sloped changes with smaller ΔP in the reference signal (Figure 30.waveform3 and waveform4), whereas waveforms containing instantaneous fluctuations (Figure 30.waveform1 and waveform2) elicited pronounced overshoots and undershoots with delayed settling times. This degradation in performance is consistent with the system constraints identified earlier: steeper pressure transitions require proportionally higher flowrates, which amplify pump-induced oscillations and introduce additional non-linearities in the outlet resistance. These effects also explain the irregular fluctuations observed near the plateau regions of the step responses, where the controller continuously attempts to compensate for the mechanical jitter and unsteady flow generated by the pump at its operating limits.

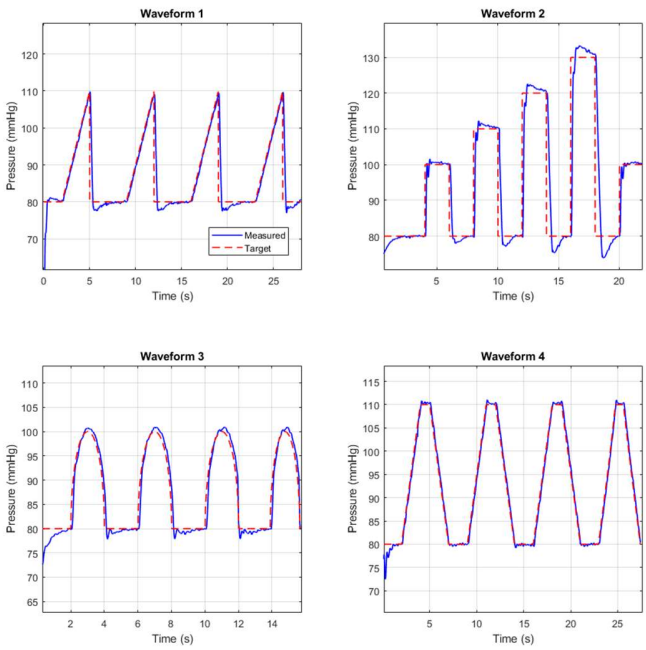


Figure 30 Controlled Input Waveform Results

The practical limitations of the system were further demonstrated by the consistent temporal lag between the projected and measured pressure profiles (Figure 31, zoomed inset). Across all stepwise tests, the reproduced waveforms exhibited a shift of Δt = 0.1 s. This latency can be attributed to the fixed sampling rate of the Deltran HX711 measurement configuration as well as the finite mechanical response of the pump, both of which limit how quickly the controller can detect and react to changes in ΔP. Consequently,

even when the PID gains are well-tuned, the system is inherently unable to initiate corrections at the exact moment the reference changes, imposing a fundamental ceiling on achievable tracking performance.

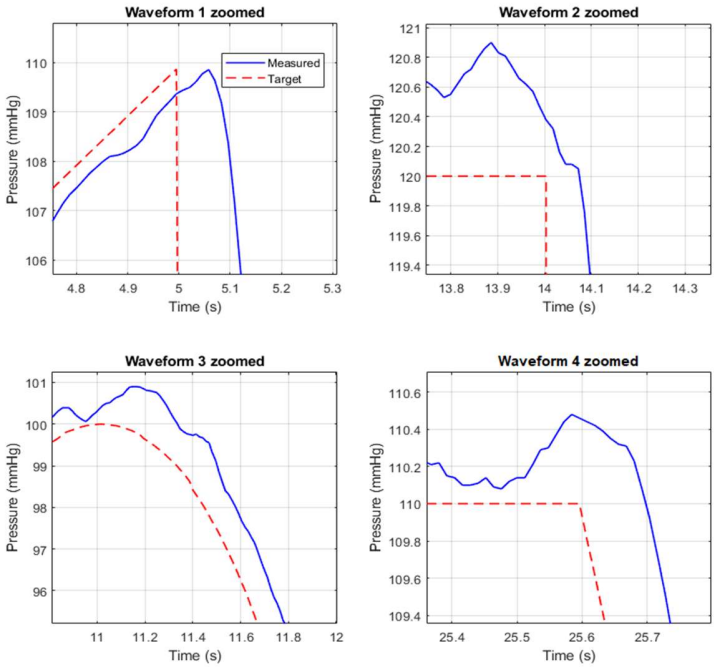


Figure 31 Controlled Input Waveform results - Zoom into peak end

3.4 Physiological Waveform Reproduction Results

The performance of the optimised pressure-tracking system under physiological waveform inputs is shown in Figures 32–34. Across the three waveform profiles (1-, 2-, and 4-second cycles), a clear trend emerges: longer waveform periods yield substantially higher fidelity, while the 1-second cycle exhibits the greatest difficulty around rapid transitions. Although the controller preserves the overall systolic–diastolic cycle duration range, the 1-second profile shows pronounced overshoot, slower tracking of steep systolic upstrokes, and increased lag during the systole-to-diastole transition. These effects stem from the system dynamics established earlier, which highlight the system limitations in regions where change in target pressure $\Delta P/\Delta t$ are the largest, this inherently causes the PID control drastically higher flowrates. When this is combined with the 77 Hz sampling limit of the sensor configuration, the PID control rapidly loses control over the reproduced waveform. This can be visually assessed by the comparing 1 second

waveforms of the same region, with all waveforms displaying unique overshoots and dicrotic notch depths.

In contrast, the 4-second waveform demonstrates markedly improved tracking performance, with minimal overshoot, reduced phase lag, and close agreement with both the systolic peak and diastolic trough. The slower rate of change in this waveform reduces the pump's dynamic demand and allows the controller to operate well within its bandwidth limits. As a result, these longer-period waveforms display far greater repeatability and produce near-identical systolic and diastolic frames between cycles, indicating more stable and reliable closed-loop behaviour.

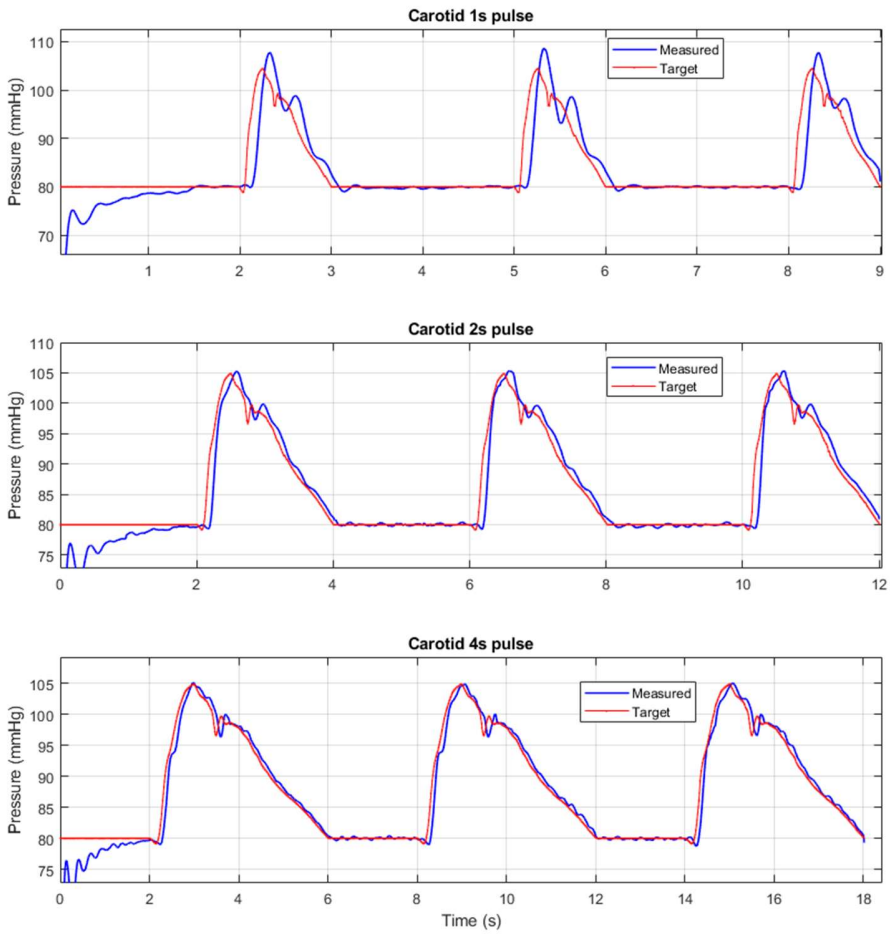


Figure 32 Physiological Waveform Reproduction for Carotid cycles

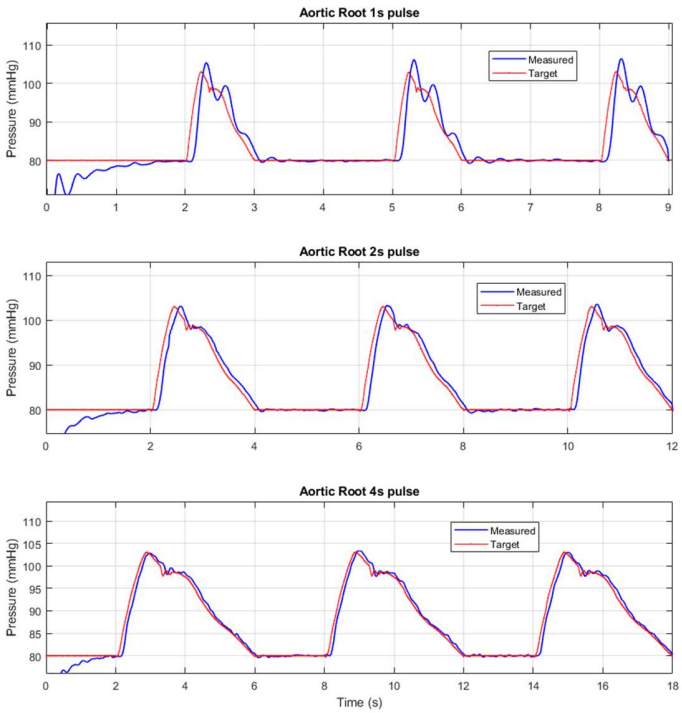


Figure 33 Physiological Waveform Reproduction for Aortic Root cycles

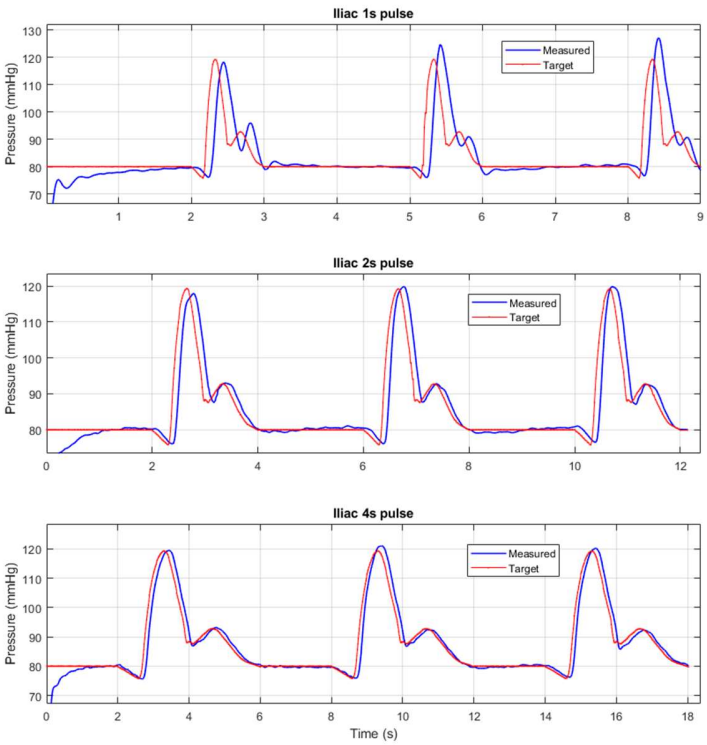


Figure 34 Physiological Waveform Reproduction for Iliac cycles

To further analyse waveform fidelity, each measured trace was aligned to its digital reference by matching the peak of the systolic upstroke, enabling direct 1:1 comparison. As shown in Figure 35, the effectiveness of the controller is strongly dependent on the required pressure rate of change ΔP . The 4-second waveforms displayed the closest agreement with the reference waveform, with minimal phase error, accurate reproduction of systolic peaks, and well-matched diastolic characteristics. The 2-second profiles also maintained acceptable fidelity, with only modest deviations but preserved overall morphology. In contrast, the 1-second cycles (Figure 35) present the largest morphological distortion, including systolic overshoot, early systolic broadening, and visible smearing of the dicrotic region. These deviations reflect the bandwidth limitations identified in earlier sections: short-period waveforms require rapid pressure modulation, which demands higher instantaneous flowrates. At these higher flowrates, pump-induced oscillations increase in amplitude, while both system inertia and the ~77 Hz sampling constraint limit the rate at which the controller can apply corrective action. As a result, the PID loop progressively loses control over the pressure transient, leading to under-damping, delayed response, and reduced shape fidelity.

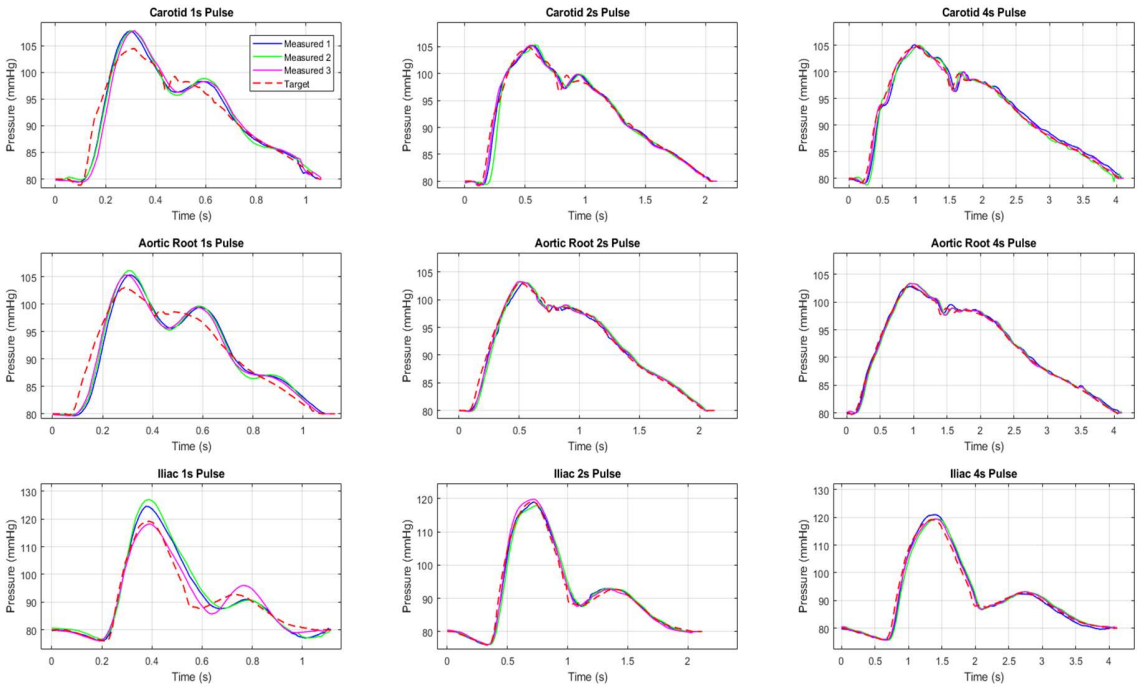


Figure 35 Superimposed Waveforms to Digital Waveforms

To more clearly illustrate how the effects of these limitations manifest across all cycle, the error between the measured pressure and its digital reference was computed for all arteries and cycle durations (Figure 36). These error traces reveal characteristic patterns that align with the qualitative observations made from the superimposed plots. For the 1-second cycles, all profiles exhibit a sharp error peak during the systolic upstroke, reaching approximately 3 mmHg for the Carotid and Aortic Root waveforms and up to 10 mmHg for the Iliac waveform. These prominent error spikes extends to other unique sections of the waveforms, as seen by the Carotid and Aortic Root 1 second graphs, which depicts largest error at sectors corresponding to points of inflection within the reference wave, more precisely the pre-systolic shoulder, and dicrotic notch.

As the waveform period increases, the tracking error decreases substantially. At 2 seconds, the majority of errors fall within ±1-2 mmHg, with minor error deviations through the pre-systolic shoulder and dicrotic notch regions. By 4 seconds, the error becomes both smaller in amplitude and more uniform across the cycle, generally remaining within ±1 mmHg across all waveform profiles.

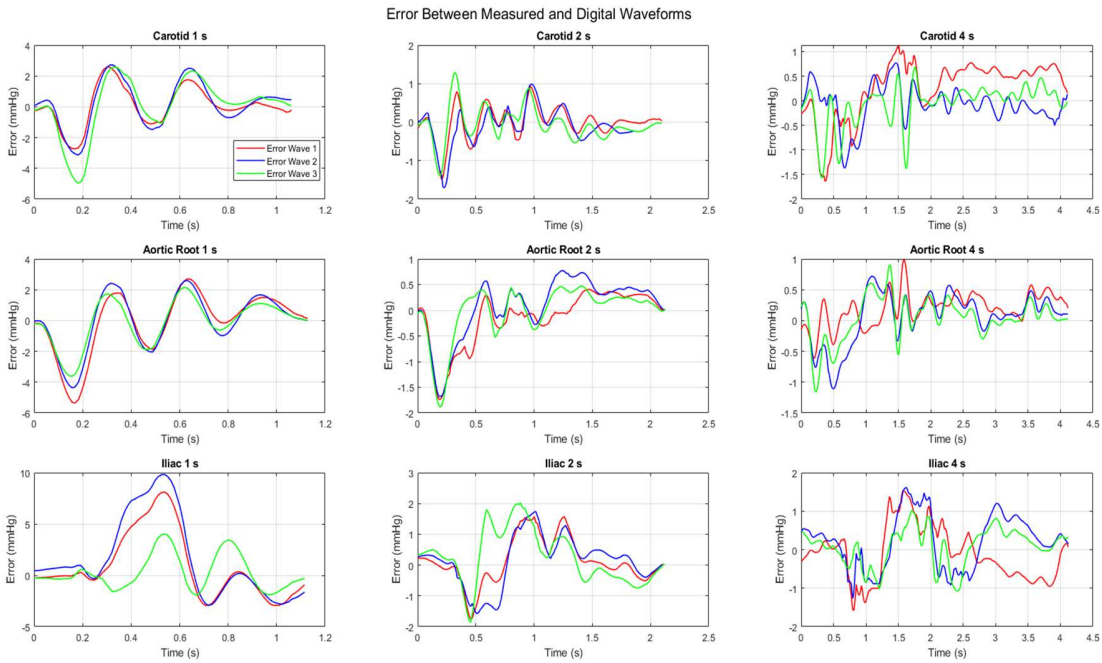


Figure 36 Error of In Vitro Waveform to Digital waveform (superimposed)

These qualitative findings are further substantiated by the quantitative metrics summarised in Table 7, which evaluates the most visually correct pulse from the 3 cycle physiological waveform procedure. Mean

Absolute Error (MAE) and Root Mean Square Error (RMSE) both decrease markedly as waveform duration increases, confirming that longer waveforms fall within the controllable bandwidth of the system. For example, the Carotid profile improves from MAE = 1.60 mmHg (1 s) to 0.36 mmHg (4 s), while the Aortic Root profile similarly reduces from 1.36 mmHg to 0.30 mmHg across the same transition. Peak Error values further highlight the influence of steep pressure gradients: the largest peak mismatches occur in the 1-second Carotid (3.32 mmHg) and Aortic Root (2.29 mmHg) cycles, consistent with the system's difficulty in following rapid systolic upstrokes. Finally, the normalised RMSE (NRMSE), which evaluates shape conformity independent of amplitude, is lowest for the 4-second waveforms (≈0.02), confirming superior morphological tracking, while the 1-second cycles present the poorest agreement (≈0.05–0.09). Taken together, these metrics quantitatively reinforce the earlier qualitative findings: waveform reproduction quality is fundamentally limited by the ΔP/Δt demands of the target waveform and the associated flowrate-dependent oscillatory behaviour of the pump.

Table 7 Best Performing Superimposed Waveform Results

Waveform	MAE_mmHg	RMSE_mmHg	PeakError_mmHg	NRMSE
Carotid 1 s	1.6073	2.4393	3.32	0.095361
Carotid 2 s	0.51177	0.7572	0.37	0.029429
Carotid 4 s	0.35612	0.56706	0.02	0.02203
Aortic root 1 s	1.3638	1.8614	2.29	0.079923
Aortic root 2 s	0.41758	0.62408	0.25	0.026796
Aortic root 4 s	0.30175	0.43063	0.36	0.018474
Iliac 1 s	1.7598	2.3428	-0.93	0.053982
Iliac 2 s	0.8578	1.2039	0.57	0.02774
Iliac 4 s	0.80594	1.2271	0.24	0.028275

3.5 Discussion

The results of this study collectively demonstrate that a low-cost, microcontroller based pressure control platform can reproduce physiological pressure waveforms with high fidelity, provided that the target waveform remains within verified controllable dynamic range for the system. The accurate replication of 2 and 4 second arterial pressure cycles, displayed in figure 36, presents the effectiveness of the implemented PID controller and suitability of the select hardware components. These findings directly address a gap in current research, where available pressure mimicking devices are predominantly higher cost and considerably larger, or smaller models which compromise on pressure tracking capabilities.

The pressure mimicking capabilities of the system establishes a foundation for future applications beyond waveform reproduction. The platform has relevance for low-cost laboratory equipment in fields such as drug delivery testing, medical device prototyping and fluid structure interaction studies, where controlled pressure environments are essential.

Despite the outlined performance observed at moderate frequencies, the physiological waveform experiments revealed several important limitations. (1) The oscillatory behaviour observed at higher flowrates, (2) Sensor sampling Bandwidth and (3) Vessel hydraulic resistance.

While the oscillatory behaviour has been attributed to the mechanical functionality of the NE-1010's linear actuator, further testing is required to verify cause of the problem. Other possibilities include disturbance of laminar flow within the vessel by any geometrical abnormality, disrupting linear resistance within its vessel. To test this directly, flowrate would need to be measured at both the outlet of the syringe and outlet for the vessel to verify the cause of oscillations. If flow disruption is caused by the Pump, possible solutions may include the use of a Piston actuated pump. This would directly eliminate the mechanical disturbances introduced by the leadscrew. However, this hypothesis must be verified experimentally through implementation and comparative testing against the existing pump.

The sensor sampling bandwidth imposed by the HX711 ADC amplifier, introduces a fixed latency that limits the system's responsiveness during rapid pressure changes, evident in figure 36. This sampling constraint directly contributes to phase lag and inability to capture higher frequency characteristic of a waveform. Using a higher speed ADC or analog front end with minimal digital filtering would significantly improve responsiveness of the system and increase resolution of resulting waveforms. Suggestions for this include using Honeywell ABP sensors, which are analog devices with a default sampling frequency of 1 kHz. It is important to note that although these sensors offer significantly higher sampling rates, integrating them into a configuration similar to the one used in this study introduces several practical challenges. These include increased priming restrictions causes by sensor's dead end geometry, more complex attachment and plumbing requirements, and exposed, uninsulated electronics, all of which make the sensor considerably more difficult to work with

The final performance bottleneck relates to the fluid resistance of the in-vitro vessel. As indicated by the curvature in figure 26, the hydraulic resistance of the current vessel is not strictly linear. Identifying or designing a vessel with more linear resistance characteristics would directly influence the effective gain applied by the proportional term (K_{p}) of the PID controller, as the error correction would remain effective

over a broader operating range. In contrast, the current non-linear flow–pressure relationship reduces the controller's effectiveness at higher pressures, where resistance increases disproportionately and the system's corrective output causes overshoot. A more linear resistance profile would therefore stabilise the pressure-flow interaction, ensuring that incremental changes in flow produce predictable pressure responses, which in turn allows the controller to maintain tighter regulation during rapid transients.

In summary, this project demonstrates that high-precision physiological waveform reproduction can be achieved using low-cost hardware, provided that each subsystem is carefully calibrated, characterised and evaluated. This platform, therefore, fills a gap by offering an accessible, modifiable, and experimentally validated tool for pressure-control research. While several constraints limit the performance at sub-second wave cycles, the modular design of the system, together with the clearly identified pathways for improvement, provide a strong foundation for future development and implementations of additional subsystems.

4 Conclusions

In this study, a low-cost, microcontroller based platform for reproducing physiological pressure waveforms was developed and experimentally validated. The systems integrates an Arduino controlled NE-1010 Syringe pump, a medical grade Deltran pressure transducer with a HX711 ADC amplifier, a custom hydraulic assembly containing a 0.6 mm ID needle outlet. This system is dynamically operated by a closed-loop PID controller, designed to mimic waveform profiles typically seen in vivo, through modulating flowrates in response to real-time pressure feedback. Three reference waveforms tested the performance of the optimised PID controller, under three varying wavelength durations of 1,2 and 4 seconds. The reproduced pressure profiles for the 2 and 4 second waveforms closely matched their digital reference in both amplitudes and morphology, producing minimal phase lag, while maintaining high fidelity as seen in Figure 36.

While sub-second pressure replication was no achieved, several improvement factors were identified. The most prominent factors affecting the performance of the system were (1) Sensor sampling bandwidth, (2) Liquid dispensing mechanism and (3) vessel hydraulic compliance. To address sensor sampling speed, sourcing a higher speed ADC amplifier would drastically improve real-time correction for more dynamic $\Delta P/\Delta t$ change. For liquid dispensing devices, linear mechanism such as piston pumps may reduce effects of oscillations. It is also important to note that further evaluation of the current pumps may be conducted to concisely determine the cause of oscillations. To configure vessel hydraulic compliance, it is recommended to develop a hydraulic model similar to that observed in Lake et al. (2017), following the geometric properties analysed in this report to more closely replicate complex cardiovascular environments and fluid dynamics.

Overall, this work demonstrates that physiologically realistic pressure waveforms can be generated using inexpensive, readily available hardware, provided that the target waveform remains within the system's characterised dynamic range. Beyond validating the feasibility of low-cost pressure regulation, the report presents a tested methodology for integrating hardware, tuning controllers, and optimising performance in microcontroller-based pressure systems. This framework offers a practical foundation for future development of low-cost pulsatile pressure platforms and establishes a repeatable approach for refining pressure-control hardware in research settings. With further refinement, the principles and methods developed here have clear practical application across a wide range of in-vitro studies and experimental environments requiring controlled, programmable pressure conditions.

5 References

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6